The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast: a factorised physics prior for surrogates

Abstract

Boussinesq-trained surrogates for thermal convection transfer poorly across temperature contrast: they learn one correction field and have no signal with which to rescale it. For steady two-dimensional buoyancy-driven enclosure convection we show that the relative contrast $\varepsilon = \Delta T/T_0$ sets the leading-order scaling of the difference between the Boussinesq and compressible solutions, while the flow sets its amplitude and spatial shape. $Fr^2 = \varepsilon$ identically, and Ma enters only at $O(Ma^2)$ in the retained low-Mach equations. Over 627 paired solutions spanning $\varepsilon = 0.02$–1.9, the relative correction is $R = A\varepsilon(1 + \kappa\varepsilon)$ with $\kappa_{\mathrm{fit}} = 0.42$–$0.83$ from a direct quadratic fit and an intercept consistent with zero. A free power law over the same window returns the window average, $m = 1.023 \pm 0.018$ and 0.995 ± 0.019 for (u_x, u_y); fitting each configuration separately over $\varepsilon \in [0.4, 1.9]$ gives a local slope of 1.17–1.45. Pressure gives $m = 1.009 \pm 0.015$ in the fixed post-processing gauge used here, and θ has no identifiable exponent. One Newton step of the low-Mach residual at the Boussinesq state reproduces the correction to 4.7/3.8/7.2% (median, $u_x/u_y/p$) at $\varepsilon = 0.02$–0.10, and the extrapolated shape predicts held-out measurements at 0.1 and 0.2 to 2.4/1.8/3.8%. The correction factorises as $D = \varepsilon P(X)$, so a surrogate learns only the spatial shape P and multiplies by ε outside the network. On a single-ε public data set this lowers the channel-averaged transfer error from 0.131 to 0.086 (1.5×) within the same boundary-condition family; across families it does not beat doing nothing. At $\varepsilon = 1.2$ the integral Nusselt number changes by 0.67% between the two members; the mean per-sample ratio is 0.54% and the wall-flux L_1 change is 4.25%; a prior trained on the same data lowers the temperature-field error from 7.2% to 0.10% and the local wall-flux error from 6.0% to 4.3% at the hot wall, but its integrated wall-Nu error there is 2.6% against 0.7% for doing nothing: the wall-normal derivative, not the field norm, is the limiting quantity. A wall-flux penalty in the same loss removes it, bringing the integrated error to 0.11% and the local error to 1.0% at $\varepsilon = 1.2$. One inference takes 6.3 ms; replacing the compressible member saves 2.9× per pair and 13× for a 20-flow × 5-contrast scan. A 15-sample 3-D consistency check ($res = 16$, $Ra = 10^4$) gives exponents within 0.16 of unity.

Keywords: natural convection, Boussinesq approximation, low-Mach compressible flow, physics-informed surrogate, neural operator, parametric dependence

Nomenclature

Roman		*Greek*	
A	prefactor of $R = A\varepsilon^m$	β	thermal expansion coefficient
D	correction $Y - X$, the surrogate target	γ	ratio of specific heats
e	internal energy	ΔT	wall-to-wall temperature contrast
E	total energy		
Fr	Froude number, $Fr^2 = \Delta T/T_0$	ε	relative contrast, $\Delta T/T_0$
$\boldsymbol{g}$	gravitational acceleration		
L	side length of the cavity	θ	normalised temperature
m	scaling exponent	κ	curvature, $R = A\varepsilon(1 + \kappa\varepsilon)$
Ma	Mach number	ρ	density
n	number of samples in a data set or split	$\boldsymbol{\sigma}$	viscous stress tensor
		χ	base-flow advection strength
Nu	Nusselt number		
p	pressure		
P, Q	first- and second-order correction fields		
$\boldsymbol{P}_{\text{work}}$	viscous work term		
Pr	Prandtl number		
q	Rayleigh exponent, $R \propto Ra^q$		
R	relative correction, $\|Y - X\|/\|Y\|$		
Ra	Rayleigh number		
Ra_{eff}	effective Rayleigh number, $Ra\,Fr^2/\varepsilon$		
Re	Reynolds number, $(Ra/Pr)^{1/2}$		
T	temperature, T_0 its reference value		
$\boldsymbol{u}$	velocity, (u_x, u_y)		
$\hat{u}$	peak Boussinesq speed		
X, Y	Boussinesq and compressible solutions	$\mathcal{L}$	linearised low-Mach operator at $O(\varepsilon)$

1. Introduction

Steady natural convection is a workhorse of thermal engineering and it is expensive. A converged pair of two-dimensional solves on a 64^2 mesh costs 34 s of wall clock on an eight-way desktop (Sec. 5.12); the three-dimensional problems that engineering actually needs cost hours to days. That is the standard motivation for surrogate modelling, and the standard response is to train a network that maps a cheap model's output to an expensive model's output [1, 2].

For convection the two models are not two arbitrary fidelities: they are the Boussinesq (Oberbeck–Boussinesq) equations and the compressible low-Mach equations. Boussinesq is exact only in the limit of vanishing relative temperature contrast; the corrections are $O(\beta\Delta T)$ and have been characterised asymptotically for decades [3, 4]. Recent work makes the pair directly learnable. Menglik et al. [5] publish paired Boussinesq→compressible steady solutions in two dimensions (4000 training pairs plus held-out splits, CC BY 4.0) and train a Fourier neural operator [6] to predict the compressible field from the Boussinesq field. That data set—which we use as an independent validation set in Secs. 5.7–5.9—contains exactly one temperature contrast: $\Delta T = 1.2$ at $T_0 = 1$.

The single-contrast design is the practical crux. A surrogate fitted on such a data set has learned the correction at one operating point. At any other contrast the Boussinesq input is unchanged—X is ε-blind—while the correction that must be

added scales with ε; the network therefore has no signal with which to rescale the correction it memorised, and the only remedy on offer is to generate a second paired data set, i.e. to pay again exactly the cost the surrogate was meant to remove. We show in Sec. 5.3 that this is not a modelling failure that more capacity or a better architecture can repair: the missing variable is ε, and we measure the resulting ceiling.

This paper makes the barrier explicit and then removes it.

What governs the gap. For the non-dimensionalisation used here the Froude number is not free: $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically, and the Mach number enters the equations only at $O(Ma^2)$ within the retained low-Mach equations (Sec. 2.2; a factor-24 change in Ma leaves the measured correction unchanged at the precision of the solve). Within the retained equations and this scaling there is therefore one free parameter rather than two, and it has a name: $\varepsilon = \Delta T/T_0$.

What the gap looks like. Expanding the compressible member in ε gives an $O(1)$ system that is term-by-term the Boussinesq system, plus an $O(\varepsilon)$ source produced by the density-modulated inertia and by the non-solenoidal velocity field $\nabla \cdot \boldsymbol{u}_1 = \nabla\cdot(\theta_0\boldsymbol{u}_0)$. Because the linearised operator is ε-independent, the correction factorises, $Y - X = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots$, with a shape P that does not rotate as ε moves (Sec. 3). Measured over two decades of ε (627 samples, Sec. 5.1) the relative correction obeys $R = A\varepsilon^m$ with $m = 1.023 \pm 0.018$ for u_x and 0.995 ± 0.019 for u_y. The pressure channel gives $m = 1.01$ in the fixed gauge of Sec. 5.2, a numerical observation rather than a gauge-invariant exponent. θ is a weak, flow-dependent channel: its free power-law fit is not identifiable ($R^2 = 0.22$; slope 1.02 over the full window, 1.67 on the 104-configuration subset), so we make no exponent claim for it; its amplitude is set by the flow, with the χ parameterisation at $R^2 = 0.78$ and the $\hat{u}$ form at 0.90 (Sec. 5.1). The exponent is window-averaged: the same 627 points give 1.23–1.32 when fitted from $\varepsilon = 0.8$ upwards, which is the ε^2 curvature. The intercept vanishes, and the prefactor's weak Rayleigh-number trend reproduces in a second geometry and in a third Prandtl number.

What it buys. Because the ε dependence can be written outside the network, a surrogate can predict $P(X)$ and multiply by ε afterwards. Supplying ε as an input as well gives the largest gain relative to the strongest conditional baseline: 1.07× in u_x for the linear form and 1.15× for the quadratic form (channel-averaged 1.09× and 1.21×; the θ and p channels are individually significant, p in the fixed gauge, while the u_x and u_y intervals include zero; Sec. 5.5)—while removing it from the input raises the extrapolation error from 0.0937 to 0.1214 in u_x and from 0.1126 to 0.1682 in p (Table 7), but that removal is what lets a network trained on a public data set that exists at a single ε be applied, without retraining, to our $\varepsilon \in [0.2, 1.9]$, where it reaches relL2 = 0.093 (u_x) against 0.144 for the plain model at the same budget (Sec. 5.9). The prior is also the cheaper model when data is scarce (Sec. 5.6: 1.4× better than an ε-augmented baseline at $n = 50$, crossover near $n \approx 300$),

the curvature it learns is positive in all four channels ($\kappa_{\text{net}} = 0.22$–$0.42$, smaller than the 0.42–0.83 of the direct quadratic fit but of the same order), and the same prior transfers to a Fourier neural operator with the same qualitative pattern (u_x unchanged, p 13% worse than with an explicit ε input).

1.1. Related work

Reduced-order models replace the discretised equations by a small number of modes; parametric dynamic mode decomposition is a representative recent example [7]. Graph convolutional networks exploit the mesh and have been applied to natural convection in annuli [8] and to geometry-adaptive steady convection with physics-informed losses [9]. Operator learning removes the mesh-resolution constraint: DeepONet approximates the solution operator from branch and trunk networks [10], physics-informed DeepONets add the residual to the loss [11], and Fourier neural operators (FNOs) learn a global convolution in Fourier space [6]. FNOs have since been used for sparse-field reconstruction [12], real-time heat-pipe startup prediction [13], and operator-level uncertainty quantification in cooling-channel optimisation [14]. Physics-informed neural networks [15] and their heat-transfer variants [16, 17] take the opposite route: they solve one instance by minimising a PDE residual rather than learning a family of instances. Reviews of the numerical analysis and of the application landscape are given by De Ryck and Mishra [18], Meng et al. [2] and Luo et al. [19].

A physics prior can enter as a residual penalty, as a conservation law built into the architecture, or as an algebraic factorisation of the output. The first two are well developed; the third is the subject of this paper. Residual-based corrector operators solve a linear variational problem for the residual of a trained operator and add the solution to its prediction; the correction is deterministic, costs one linear solve per query, and improves accuracy without retraining the base operator [20, 21]. Physics-informed extrapolation of deep operators has been demonstrated by adding sparse observations or residual terms [22], and transfer operator learning addresses distribution shift between parameter regimes [23]. Multi-fidelity surrogates combine cheap and expensive data through LSTM corrections [24], feature-adjacent physics-informed learning [25] or transfer-learned FNOs [26]. These methods reduce the amount of high-fidelity data, but they still learn the parameter dependence from samples. For buoyant convection, the Oberbeck–Boussinesq approximation and its validity limits are classical [3, 4], and the review of Ahlers et al. [27] summarises the heat-transfer and large-scale dynamics of turbulent Rayleigh–Bénard convection. Non-Oberbeck–Boussinesq (NOB) effects have been measured under strong temperature contrast [28], in cryogenic helium [29] and in a water-filled differentially heated cavity [30]. At $Pr = 0.71$ and large contrast they are not a small correction: they grow cold-corner vortices and can reverse the large-scale circulation [31]. On the numerical side, low-Mach compressible formulations for enclosed thermal convection have been developed and compared with the Boussinesq limit [32, 33], and the rigorous derivation of the Oberbeck–Boussinesq approximation has recently

been revisited [34]. Bella et al. obtain the Oberbeck–Boussinesq system with a non-local boundary condition for the temperature deviation in the low-Mach limit of the compressible Navier–Stokes–Fourier system with Dirichlet temperature conditions; that boundary condition belongs to the full compressible-to-Oberbeck–Boussinesq limit, whereas the expansion here is taken in the retained low-Mach system with fixed wall temperature, where the temperature deviation vanishes at every order on the walls. What is missing is a quantitative, closed-form account of how the *difference between the two solutions* scales with the temperature contrast, and a surrogate design that uses that scaling. These studies fix ε and compare the two solutions; none extracts the ε dependence as a reusable prior. The present work differs in that the parameter dependence is derived, and only the spatial shape of the correction is learned.

Table 1 places the resulting prior among the four families.

Contributions.

1. We show that, in the non-dimensionalisation used here and within the retained low-Mach equations, the Boussinesq–compressible gap is controlled by one parameter, $\varepsilon = \Delta T/T_0$, rather than by ε and Ma independently (Sec. 2.2).
2. We derive the correction's factorisation from an ε-expansion and verify all five of its predictions numerically, including the divergence relation (Sec. 3).
3. We calibrate the resulting law on 627 paired samples spanning two decades of ε in the unit square at $Pr = 0.71$, test it in a second geometry (104 samples) and at $Pr = 4.38$ (104 samples), and validate the prefactor on an independently generated public data set—including the secondary Rayleigh-number trends (Secs. 5.1, 5.7, 5.10, 5.13).
4. We measure what the gap does to the wall heat transfer: at $\varepsilon = 1.2$ the integral Nusselt number moves by 0.67% (ratio of ensemble means; per-sample mean 0.54%, median 0.58%) on the public data, while the local wall flux is redistributed by 4.25%; on the five isothermal-sidewall configurations the integral change at $\varepsilon = 1.9$ is -5.1% at $Ra = 10^4$ and -1.0% at $Ra = 10^3$. The 15 random-wall configurations are not used for a relative Nu statistic: eight of them have $|Nu_X| \leq 1$, where the ratio is dominated by its denominator (Sec. 5.8).
5. We turn the law into a surrogate design rule ($D = \varepsilon P(X)$), measure what it is worth (1.09× lower channel-averaged error than the strongest conditional baseline in extrapolation, with θ and p individually significant (p in the fixed gauge) and the u_x/u_y intervals including zero, and indistinguishable within seed scatter in interpolation), and show it makes an existing single-ε public data set reusable, with the end-to-end saving measured per pair (2.9×), for the 20-flow × 5-contrast scan (13×) and for a single flow swept over 100 contrasts (2.3×10^2) (Secs. 5.4–5.9, 5.12).

All results are two-dimensional except the fifteen-sample, five-contrast three-dimensional check of Sec. 5.14. Section 6 states the measured cost of the three-

dimensional problem on the same hardware and gives the concrete reasons we do not claim a 3-D sweep.

Table 1: Where the present prior sits among physics-informed operator methods. "Derived" is obtained in closed form from the equations; "learned" is supplied by the network. Rows: post-hoc correctors [20, 21], physics fine-tuning and multifidelity [22], transfer operator learning [23], dimension-informed operator [35].

family	derived	learned	target data	per query	law
post-hoc correctors	linear residual	none	no	one linear solve	no
physics fine-tuning	residual / data	correction	yes	no	no
transfer operator	none	operator	yes	no	no
dimension-informed	equivariance	operator	no	no	no
this work	ε expansion	shape P	no	no	m, κ

2. Problem, discretisation and the break-symmetry parameter

2.1. Two members per sample

Every sample is a *paired* steady solution of one physical configuration with two equation sets on the same mesh ($res = 64$, $L = 1$, $Pr = 0.71$, $\gamma = 1.4$). The high-fidelity member Y is the compressible low-Mach formulation in $\boldsymbol{w} = (\boldsymbol{u}, T, \rho, p)$:

$$\nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{1}$$

$$\nabla \cdot (\rho\, \boldsymbol{u} \otimes \boldsymbol{u}) = -\nabla p + \frac{1}{Re}\nabla \cdot \boldsymbol{\sigma} - \frac{1}{Fr^2}\rho \boldsymbol{g}, \tag{2}$$

$$\nabla \cdot (\rho E \boldsymbol{u}) = \frac{\gamma}{Re\, Pr}\nabla^2 e + \nabla \cdot (\boldsymbol{P}_{\text{work}}\boldsymbol{u}) + O(Ma^2), \tag{3}$$

$$\rho e = \gamma Ma^2 p + 1, \qquad Re = (Ra/Pr)^{1/2}. \tag{4}$$

The low-fidelity member X is the Boussinesq system in $(\boldsymbol{u}, T, p)$, written in the *same* free-fall scales:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{5}$$

$$\boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \left(\frac{Pr}{Ra}\right)^{1/2} \nabla^2 \boldsymbol{u} + \theta \hat{\boldsymbol{g}}, \tag{6}$$

$$\boldsymbol{u} \cdot \nabla T = (Ra\, Pr)^{-1/2}\nabla^2 T, \qquad \theta = \frac{T - T_0}{\Delta T}. \tag{7}$$

Both are solved with Taylor–Hood-type mixed elements, damped Newton iterations and a MUMPS factorisation; steady states are reached by an adaptive march with a convergence guard. Both members of a pair start from the same initial field and the same wall-temperature seed, so they follow the same branch. The temperature boundary data are random Fourier wall temperatures with per-wall mode dials; the classical differentially heated cavity (constant hot and cold side walls, adiabatic top

and bottom) is the one-mode member of the same family. Both are buoyancy-driven enclosure flows, and the results below are not restricted to the Rayleigh–Bénard layer.

The generator draws Ra and ε independently. At fixed Ra, changing ε changes $Fr = \sqrt{\varepsilon}$, i.e. the buoyancy scale, while $Re = (Ra/Pr)^{1/2}$, Pr and γ stay fixed. The Boussinesq member therefore sees only Ra and the thermal boundary data; the compressible member sees the same flow at a different Fr. This is a statement about the dimensionless parameterisation, not a claim that one fixed apparatus realises every pair: at fixed Ra, changing ε requires the dimensional grouping $g\beta L^3/(\nu\alpha)$ to change in inverse proportion to ΔT. A fixed apparatus follows the one-dimensional slice $Ra \propto \varepsilon$; the (Ra, ε) rectangle contains that slice but is not restricted to it.

2.2. Fr is not a parameter, Ma is negligible, ε is the axis

Table 2 lists the three candidates for a break-symmetry axis and what rules each one out.

Table 2: The three candidate break-symmetry parameters. Only ε survives as a free axis in the present non-dimensionalisation. The Ma row is a consistency check on the retained equations, which drop the $O(Ma^2)$ term.

candidate	evidence	conclusion
Fr	the generators state $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically; the released 2-D code pinned $Fr = 1.095$, correct only at $\Delta T = 1.2$	not free; pinned-Fr sweeps of $T_0/\Delta T$ are an artefact
Ma	the generator sets $Ma = 1.78 \times 10^{-3}(Ra/10^6)^{1/6}$, i.e. 3.8×10^{-4} at $Ra = 10^2$ and 1.8×10^{-3} at $Ra = 10^6$; fixing Ra, T_0, ΔT and changing Ma by 24× leaves the measured correction unchanged to a relative 10^{-5}	$O(Ma^2)$ is dropped by the retained equations; the sampled range does not resolve it
$\varepsilon = \Delta T/T_0$	same boundary-condition seed stream, ΔT swept	the clean, one-dimensional break-symmetry axis

2.3. The low-fidelity solution is exactly ε-blind

θ, Pr and Ra are the only quantities that appear in the Boussinesq weak form; neither T_0 nor ΔT appears alone. Hence, with the thermal data prescribed in θ and Ra fixed, X cannot depend on ε. This was verified to solver precision: across 20 configurations × 5 values of ε (same Ra, bit-identical θ boundary stream), the maximum relative deviation of X is 9.77×10^{-8}.

> **Corollary.** The amplitude of the correction is set by ε, which is *absent from the surrogate's input*. The spatial shape of the correction is a function of X and is learnable; no function of X alone, however expressive, can supply the factor ε that scales it. The limit is informational, not architectural (Sec. 5.3).

3. Asymptotic theory: why the correction factorises as $\varepsilon P(X)$

Write $T = T_0(1 + \varepsilon\theta)$ with $\theta = O(1)$. With $\rho = 1/T$ (the equation of state and initial condition used here) and $Fr^2 = \varepsilon$, the buoyancy source is

$$\frac{\rho \boldsymbol{g}}{\varepsilon} = \frac{\boldsymbol{g}}{\varepsilon} - \theta \boldsymbol{g} + \varepsilon\theta^2 \boldsymbol{g} + \ldots, \tag{8}$$

the constant $\boldsymbol{g}/\varepsilon$ is balanced by a hydrostatic pressure p_{-1}/ε with $\nabla p_{-1} = -\boldsymbol{g}$, which drops out at the next order, so the $O(1)$ balances are

$$\nabla \cdot \boldsymbol{u}_0 = 0, \tag{9}$$

$$\boldsymbol{u}_0 \cdot \nabla \boldsymbol{u}_0 = -\nabla \tilde{p}_0 + Re^{-1} \nabla^2 \boldsymbol{u}_0 + \theta_0 \hat{\boldsymbol{g}}, \tag{10}$$

$$\boldsymbol{u}_0 \cdot \nabla \theta_0 = (Re\, Pr)^{-1} \nabla^2 \theta_0, \tag{11}$$

which is term-by-term the Boussinesq system solved as X ($Re^{-1} = (Pr/Ra)^{1/2}$). The correction at $O(\varepsilon)$ is driven by the density modulation:

$$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0), \tag{12}$$

$$\nabla \cdot (\boldsymbol{u}_1 \otimes \boldsymbol{u}_0 + \boldsymbol{u}_0 \otimes \boldsymbol{u}_1 - \theta_0 \boldsymbol{u}_0 \otimes \boldsymbol{u}_0) = -\nabla \tilde{p}_1 + Re^{-1} \nabla^2 \boldsymbol{u}_1 + \theta_1 \hat{\boldsymbol{g}} - \theta_0^2 \hat{\boldsymbol{g}}. \tag{13}$$

The energy equation contributes at the next order. Writing $T = T_0(1 + \varepsilon\theta_0 + \varepsilon^2\theta_1 + \ldots)$, the $O(\varepsilon)$ balance is the base energy equation for θ_0 already contained in $\mathcal{L}$, and θ_1 enters at $O(\varepsilon^2)$ in T; the weight $1/Fr^2 = 1/\varepsilon$ in (8) promotes that same term to $O(\varepsilon)$ in the momentum balance, which is why θ_1 appears in (13). The relative correction in the normalised θ is still $O(\varepsilon)$, because it is measured against θ_0, whose norm is the largest of the four fields; that normalisation, not a higher order in θ_1, is why θ has the smallest relative amplitude. The $O(\varepsilon)$ system is linear, with coefficients fixed by X. Its mass equation is inhomogeneous, so the operator is not the Boussinesq operator linearised about X—whose velocities are divergence-free—but the linearisation of the retained low-Mach system about X, with the mass constraint carried inside the operator. We denote that operator by $\mathcal{L}$; its inverse, up to the constant-pressure gauge, is written $\mathcal{L}^{-1}$. It is solvable: its only singular mode is the constant pressure, and the compatibility condition $\int \nabla \cdot (\theta_0 \boldsymbol{u}_0)\, d\Omega = \oint \theta_0 \boldsymbol{u}_0 \cdot \boldsymbol{n}\, dS = 0$ holds because $\boldsymbol{u}_0 \cdot \boldsymbol{n} = 0$ on the walls, so $\tilde{p}_1$ is determined up to the gauge. Since $\mathcal{L}$ is built from X and the retained equations, and contains no explicit ε,

$$D = Y - X = -\varepsilon \mathcal{L}^{-1} S_1[X] - \varepsilon^2 \mathcal{L}^{-1} S_2[X] + \cdots = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots \tag{14}$$

Five predictions follow: $R \to 0$ as $\varepsilon \to 0$; $R \propto \varepsilon$ with a vanishing intercept; the shape of D does not rotate with ε; $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$; and the ε^1 term carries most of the correction energy. All five hold; Table S1 of the supplementary material lists them and their measured values. Table S1 also measures the remainder: the ε^1 term carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, and $\varepsilon^1 + \varepsilon^2 + \varepsilon^3$

carries 99.9%, so the two-term truncation leaves 1.0% over the sampled range. This is a measured statement about the retained low-Mach system, not a proof of convergence.

The divergence identity is not exact at the top of the range: the cosine between its two sides falls from 0.994 to 0.901 as ε rises to 1.9, which is the footprint of the ε^2 term. The fifth row of Table S1 quantifies that term independently.

Domain of validity. $\varepsilon < 2$, since the cold wall $T_c = T_0 - \Delta T/2$ must be positive. At $\varepsilon = 1.9$ the cold wall is $T_c/T_0 = 0.05$, so the density expansion is close to its limit.

4. Data sets and protocols

Table 3 summarises the data sets used below: a 600-sample random-boundary scan calibrates the law and trains the surrogates, a 20×5 paired scan isolates ε, and the public, geometry and fluid sets supply the external checks.

Table 3: Data sets used in this study. The public sets are from Menglik et al. [5].

set	n	what varies	purpose
random-BC	600	$Ra \sim \log U[10^2, 10^6]$, $\varepsilon \sim U[0.2, 1.9]$, 134 wall-mode families	calibrate the law, train surrogates
$\varepsilon \to 0$ checks	27	$\varepsilon = 0.02/0.05/0.10 \times Ra = 10^3/10^4/10^5$, three draws each	anchor $R(0) = 0$ (included in the fit)
paired	20×5	only ε; Ra and the θ boundary stream fixed within a configuration	ε-blind reference, shape invariance
aspect 2:1	104	same law, geometry 2×1	external validity
tight rerun	10	the 10 worst-converged samples, tolerance $10^{-6} \to 10^{-8}$	noise control
public, random walls [5]	$4000 + 2 \times 200$	$\varepsilon \equiv 1.2$; two held-out splits differ in the range of the wall dials	external validation, cross-data-set transfer
public, classical cavity [5]	$5000 + 200$	constant hot/cold side walls, adiabatic top and bottom, $\varepsilon \equiv 1.2$; its 200 held-out pairs are the third split above	external validation of the prefactor
$Pr = 4.38$ (this work)	104	$Pr = 4.38$; BC/Ra/ε draws bit-identical to 104 of the random-BC samples	test of the fluid axis

Surrogates use residual learning $\hat{Y} = X + \mathcal{N}(\cdot)$, a U-Net of width 32 (or an FNO2d with 12 modes, width 32, four layers, 1.19 M parameters), Adam with cosine annealing, 200 epochs, batch 16, and targets normalised per channel. Inputs and targets are scaled per channel by standard deviations of the training split; in the cross-data-set transfer experiment the target pool is scaled by the standard deviation of its own inputs, which uses no labels. Two splits are used throughout: extrapolation (train $\varepsilon \le 1.3$, test $\varepsilon > 1.3$, $n = 218$) and interpolation (stratified 2:1

in ε, $n = 200$). Errors are the per-channel relative L_2 error against Y; "floor" means $\hat{Y} = X$.

Tables that support the argument but are not needed in its flow—the asymptotic predictions, the two-parameter fits, the exact single-ε transfer matrix, the learning curve, the public data splits, the prefactor–velocity law and the map from scripts to reported numbers—are collected in the supplementary material; they are cited where they are used.

Temperature normalisation. Eq. (7) defines the zero-mean variable $\theta = (T - T_0)/\Delta T$. The public release and our own pipeline store T/T_0 and evaluate every temperature error on the wall-normalised $\theta_w = (T - T_c)/\Delta T \in [0, 1]$, which differs from θ by the constant $1/2$. All R_θ and A_θ quoted below are in the wall-normalised convention. The same field difference reads 0.0220 wall-normalised against 0.0869 zero-mean on the public random-wall test split, and 0.0836 against 0.1628 on the classical cavity; the two conventions must not be mixed. In Table 4 the θ amplitude is therefore listed in both, 0.010 wall-normalised and 0.034 zero-mean.

5. Results

5.1. The ε-law, calibrated over two decades

Fig. 1 shows the collapse of u_x, u_y and p. The θ panel scatters about the fitted line ($R^2 = 0.22$), because its amplitude also depends on the flow (χ or $\hat{u}$); Table 4 collects the fitted exponents and amplitudes. The quoted ± is the standard error of the log–log fit. On the 600-sample scan alone the fit gives $m = 1.08$ (u_x) and 1.00 (u_y); resampling its 134 wall-mode families (2000 cluster bootstraps) gives 95% intervals [1.03, 1.14] and [0.94, 1.06], against [1.03, 1.14] and [0.95, 1.05] under an i.i.d. bootstrap. Clustering by boundary family adds no detectable uncertainty to the exponent, and the shift from 1.08 to 1.02 between the two fits is the effect of the 27 low-contrast points, not of the uncertainty.

Table 4: Power-law fit $R = A\varepsilon^m$ over the full two-decade window (627 samples). The intercept is fitted separately on the small-ε subset ($\varepsilon \leq 0.5$). For θ the amplitude is given in both conventions, wall-normalised/zero-mean. The p row uses the fixed gauge and hydrostatic correction of Sec. 5.2; its exponent is not gauge-invariant. The θ row is reported for completeness; its slope is subset-dependent and no exponent is claimed.

channel	$R = A\varepsilon^m$	A	R^2	intercept ($\varepsilon \leq 0.5$)
u_x	$m = 1.023 \pm 0.018$	0.147	0.84	$+0.0003 \pm 0.0047$
u_y	$m = 0.995 \pm 0.019$	0.193	0.82	$+0.0030 \pm 0.0062$
θ	$m = 1.024 \pm 0.077$	0.010/0.034	0.22	-0.0003 ± 0.0012
p	$m = 1.009 \pm 0.015$	0.148	0.87	$+0.0016 \pm 0.0030$

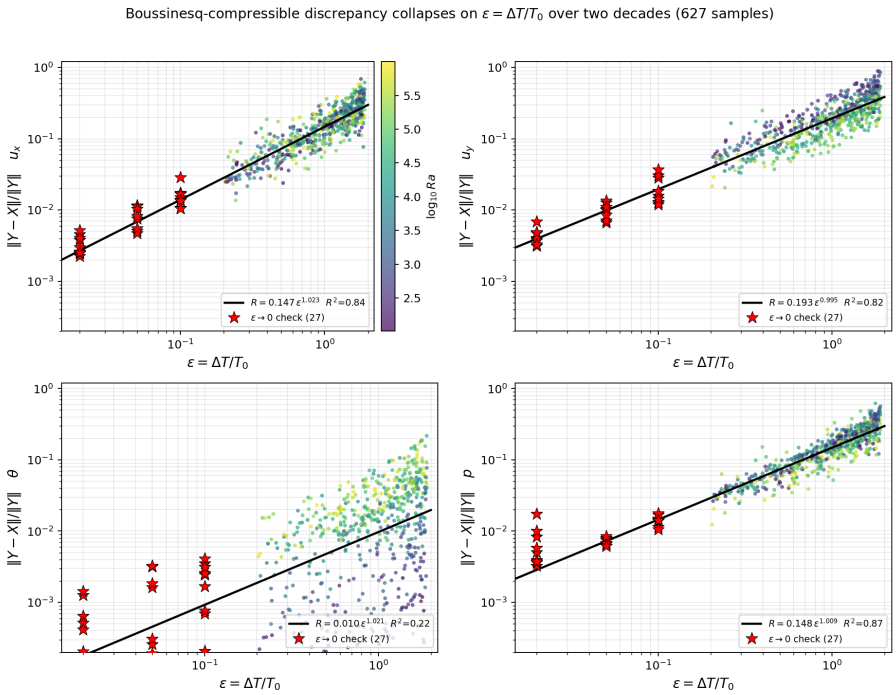


Figure 1: The Boussinesq–compressible discrepancy collapses on $\varepsilon = \Delta T/T_0$ over two decades for u_x, u_y and p; the θ panel scatters ($R^2 = 0.22$) and collapses on the peak Boussinesq speed $\hat{u}$ (Fig. 5), while a separate χ parameterisation gives $R^2 = 0.78$ alone (Sec. 5.1). Points: 600 random-boundary-condition samples, coloured by $\log_{10} Ra$; stars: the 27 small-ε checks ($\varepsilon = 0.02$–0.1, three Rayleigh numbers, three independent wall draws each), included in the fit to anchor the $\varepsilon \to 0$ end and not used as a holdout. Lines: the power-law fits of Table 4.

The exponent is window-averaged, and the curvature is real. The fit above uses the full two decades, including the 27 check points at $\varepsilon \leq 0.1$ where R is closest to its linear limit; these points are in the fit, not a holdout. Fitting only $\varepsilon \geq 0.8$ gives $m = 1.23 \pm 0.08/1.25 \pm 0.08/1.21 \pm 0.34/1.32 \pm 0.06$, and fitting each of the 20 paired configurations separately over $\varepsilon \in [0.4, 1.9]$ gives a median $m = 1.31/1.17/1.21/1.45$ (median $R^2 = 0.98$). The 104 paired configurations used for the Prandtl comparison (Sec. 5.13) give $1.16/1.05/1.67/1.11$ over their full range; the θ slope is the least stable because its signal is the weakest ($R^2 = 0.22$ here and 0.33 in that subset), and it moves much more than the velocity and pressure slopes. That is the signature of $R = A\varepsilon(1 + \kappa\varepsilon)$ rather than of an exact power: a power law fitted to a curved function over a window that excludes small ε returns $m \approx 1 + \kappa\langle\varepsilon\rangle$. The quadratic fit $R = a\varepsilon + c\varepsilon^2 + b$ gives $c > 0$ in all four channels (u_x: $+0.0513 \pm 0.0133$, 3.9σ; u_y: $+0.0695$; θ: $+0.0060$; p: $+0.0539$) with $\kappa_{\text{fit}} = c/a = 0.70/0.83/0.42/0.73$ (here a and c are coefficients of this quadratic fit, not the power-law A of Table 4). We therefore state the law as first order in ε, $R = A\varepsilon(1 + O(\varepsilon))$, quote $m = 1.02$ as

the number a wide-window fit returns, and note that the network recovers the ε^2 curvature with the same sign and order of magnitude (Sec. 5.5, $\kappa_{\mathrm{net}} = 0.22$–$0.42$; P2 fits amplitude and curvature jointly, so κ_{net} is 1.3–2.6 times smaller than κ_{fit} but has the same sign in every channel) and by the mode expansion (Sec. 3, Table S1 of the supplementary material: $\varepsilon^1 + \varepsilon^2$ carry 99.0% of the correction energy).

Mesh independence. One paired configuration (the classical cavity at $Ra = 10^4$) was rerun at 128^2 over the same five contrasts. The relative correction R changes by at most 0.1% in u_x and u_y, 0.7% in θ and 1.1% in p. Fitted over $\varepsilon \in [0.4, 1.9]$, the exponents are 1.192/1.208/1.247/1.305 at 64^2 and 1.192/1.208/1.247/1.300 at 128^2 ($u_x/u_y/\theta/p$), so the refinement moves the exponent by at most 0.005. For this configuration the exponent is not set by the 64^2 discretisation.

At the top of the Rayleigh range we repeated the check for a constant-wall configuration at $Ra = 10^6$, over $\varepsilon = 0.4$, 1.2 and 1.9. The exponents are 1.067, 1.129, 1.301 and 1.335 at 64^2, against 1.063, 1.124, 1.289 and 1.321 at 128^2 (for u_x, u_y, θ and p), a shift of at most 0.015. The largest pointwise change in R is 3.3% (p at $\varepsilon = 1.9$); the other channels change by 1.0–1.9%. The thermal boundary layer at $Ra = 10^6$ is $\approx 0.032L$ thick; the 64^2 mesh places three temperature and five velocity nodes across it, the 128^2 mesh five and nine. The 128^2 values are the reference for the exponent and pointwise checks: the 64^2 exponent is accurate to 0.015, and an individual $\varepsilon = 1.9$ correction carries a discretisation uncertainty of up to 3.3%. A 192^2 pair at $\varepsilon = 1.9$ and a 256^2 pair at $\varepsilon = 1.2$ set the level of the integral correction.

At $Ra = 10^5$ the same isothermal-wall check at $\varepsilon = 1.2$ and 1.9 gives a wall-Nu change of -0.15 and -0.36 percentage points between 64^2 and 128^2 (Nu_X itself moves by 0.86%); the integral correction moves from +0.87% to +0.73% at $\varepsilon = 1.2$ and from -1.81% to -2.16% at $\varepsilon = 1.9$. Across $Ra = 10^4$, 10^5 and 10^6 the change between 64^2 and 128^2 is therefore at most 0.38 percentage points, and the sign and order of the integral correction are unchanged.

At $Ra = 10^6$ the isothermal-wall pair was refined to 192^2 and, at $\varepsilon = 1.2$, to 256^2. At $\varepsilon = 1.2$ the wall-Nu correction falls from +0.82% at 64^2 to +0.56% at 128^2 and +0.44% at 256^2, where eight temperature nodes span the thermal boundary layer. At $\varepsilon = 1.9$ it moves from -1.50% at 64^2 to -1.88% at 128^2 and -2.10% at 192^2. Both sequences are still moving at the finest mesh, by 0.12 and 0.22 percentage points per step, so the integral values carry a resolution uncertainty of that order; the sign and the order of magnitude do not change. The $\varepsilon = 1.9$ pair at 256^2 did not reach the steady tolerance within the solver's iteration budget, at pseudo-time steps 0.05 and 0.025.

- For u_x, u_y and p the exponent is consistent across geometry within its uncertainty (0.7–1.8 standard errors; p in the fixed gauge); the prefactor A is not (0.147 $\rightarrow$ 0.175 for u_x, 0.148 $\rightarrow$ 0.222 for p, 0.193 $\rightarrow$ 0.175 for u_y, 0.010 $\rightarrow$ 0.008 for θ).

- Boundary-condition families barely matter *inside this family*: 134 wall-mode

families, 133 free dummies. Adding them to the ε-only fit raises the adjusted R^2 by 2.3 pp (p), 2.9 pp (u_y), 3.3 pp (u_x) and 8.5 pp (θ)—the ε-scaling is not an artefact of which wall modes were drawn.

- The θ channel needs one extra, cheaply available parameter, the advection strength of the base flow $\chi = \|\boldsymbol{u}\|\|\nabla\theta\|/\|\theta\|$: $R_\theta = \varepsilon(-0.0037 + 0.226\chi)$ with the ε factor imposed, not fitted; R^2 0.185 $\rightarrow$ 0.778 (Spearman 0.902), i.e. $R_\theta \approx 0.23\varepsilon\chi$. This is a parameterisation of the θ amplitude, not an independent measurement of its exponent. For $u_x/u_y/p$ adding χ does nothing. A single-input form of the same statement, $R_\theta/\varepsilon = c\hat{u}^a$, is calibrated on two independent generators in Sec. 5.7 and carries the same information with $R^2 = 0.90$.

- The fitted θ amplitude is not predictive: $A_\theta = 0.010$ with $m = 1.024$ gives 0.012 at $\varepsilon = 1.2$, while the median R_θ on the public data is 0.0228. The factor 1.9 is the 78% unexplained variance of the θ fit, and is a further reason no θ exponent is claimed.

- The pressure channel required a gauge and hydrostatic correction before the two members could be compared (Sec. 5.2).

5.2. The pressure channel: gauge and hydrostatic alignment

The two members store pressure in different conventions. The compressible raw pressure carries a hydrostatic ramp 12× larger than its dynamic part, and the two weak forms use opposite sign conventions. Comparing raw pressures therefore measures the conventions rather than the flow: the per-sample median of $|X_p|/|Y_p|$ is 0.065, that of $\cos(X_p, Y_p)$ is -0.105, and the p-channel "law" collapses to a constant ($R_p \approx 1.0$). We align the members with the protocol used by the public data set—subtract the mean, remove a least-squares linear-in-y fit, and flip the sign of the Boussinesq member. Under the same median convention the aligned pair gives 0.960 and cos = +0.991, against 0.929 and +0.986 for the public training set, so the two pairings agree to within 3.5%. The channel then follows the same law as the velocities ($A = 0.148$, $m = 1.009$). All p-channel numbers in this paper use this gauge. Pressure is gauge-dependent: the exponent is a statement about this fixed post-processing gauge, not a gauge-invariant law.

5.3. The ε-blind ceiling, measured exactly

Within a configuration X is identical for all ε, so an ε-blind model must emit one field F for every ε it is applied at (Fig. 2, Table 5). Two reference protocols are computed directly from the paired data rather than fitted: (A) train at the one ε a publicly released data set offers (here $\varepsilon = 1.2$) and apply it everywhere, error $\|Y(\varepsilon) - Y(1.2)\|/\|Y(\varepsilon)\|$; (B) one field chosen without ε, the unweighted ε-mean of Y. (B) is a compromise fixed by that choice of average, not a per-ε lower bound—a field tuned to $\varepsilon = 0.8$ gives 0.067 at $\varepsilon = 0.4$, better than (B)'s 0.180 there, and

below $\varepsilon \approx 0.8$ doing nothing beats both. Median over 20 configurations, channels $u_x/u_y/\theta/p$:

Table 5: An ε-blind reference. Protocol (A) is what a single-ε public data set forces; protocol (B) is the unweighted ε-mean field, a reference rather than an optimum for the relative L_2 error. Entries are medians over 20 configurations, ordered $u_x/u_y/\theta/p$; every entry is a relative L_2 error with $\|Y\|$ in the denominator, the same convention as the surrogate errors.

ε	do nothing ($\hat{Y} = X$)	(A) train at $\varepsilon = 1.2$ only	(B) ε-mean field
0.4	0.060/0.098/0.009/0.060	0.149/0.229/0.021/0.159	0.180/0.255/0.023/0.196
0.8	0.124/0.198/0.019/0.123	0.081/0.121/0.011/0.086	0.107/0.141/0.013/0.134
1.2	0.200/0.307/0.031/0.201	0/0/0/0	0.035/0.036/0.003/0.053
1.6	0.292/0.440/0.045/0.298	0.110/0.144/0.014/0.121	0.077/0.117/0.012/0.080
1.9	0.388/0.575/0.060/0.424	0.243/0.295/0.031/0.273	0.209/0.268/0.028/0.235

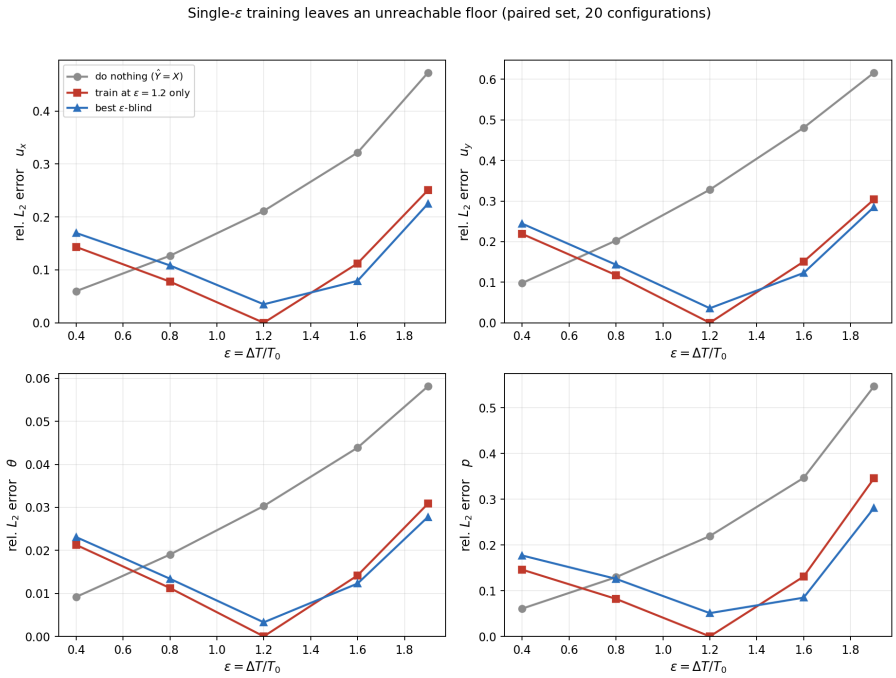


Figure 2: Single-ε training leaves an ε-blind floor. Curves show "do nothing", training at $\varepsilon = 1.2$ only, and the unweighted ε-mean field, for the 20 paired configurations.

Protocol (A) is what a single-ε public data set forces: 24% (u_x), 30% (u_y), 3.1% (θ) and 27% (p) error at $\varepsilon = 1.9$, 8% (u_x) already at $\varepsilon = 0.8$, and—in practice—at $\varepsilon = 0.4$ "do nothing" is *better* than it, and the two cross between $\varepsilon = 0.4$ and 0.8. Protocol (B), the unweighted mean over the range, still leaves 20.9/26.8/2.8/23.5% ($u_x/u_y/\theta/p$) at $\varepsilon = 1.9$. The whole effect is one line of arithmetic, and the paired set measures it exactly (the diagonal is identically zero because the same ε is the same

solution): the full 5×5 matrix for u_x is Table S3 of the supplementary material.

Within a paired configuration the error is set by the distance in ε—which is precisely the quantity $\varepsilon P(X)$ puts back into the model.

The shape of the correction is stable even though its amplitude is not. Over the 20 paired configurations, the median cosine similarity between $D(\varepsilon)$ and $D(1.2)$ stays above 0.952 over $\varepsilon = 0.4$–1.9, with minima 0.974 (u_x), 0.976 (u_y), 0.982 (θ) and 0.952 (p; Table 6). The largest deviation from unity is 0.048; a scalar amplitude therefore carries the contrast dependence, as Eq. (14) assumes.

Table 6: Cosine similarity between the correction field $D(\varepsilon)$ and $D(1.2)$, median over the 20 paired configurations. Values are ordered by ε.

channel	0.4	0.8	1.2	1.6	1.9
u_x	0.978	0.994	1.000	0.992	0.974
u_y	0.993	0.998	1.000	0.996	0.976
θ	0.991	0.997	1.000	0.996	0.982
p	0.964	0.991	1.000	0.990	0.952

5.4. Network baselines: the missing input is a scalar, not capacity

Table 7: Surrogate ablation on the two fixed splits (U-Net, width 32, 200 epochs, mean of three seeds; largest standard deviation over all cells 0.0051). A sees only the four Boussinesq fields; B also receives ε as a constant input channel; P1 multiplies ε outside the network but still reads it as an input; P2 adds a learned quadratic factor; P3 removes ε from the input entirely. Errors are per-channel relative L_2 errors against Y. The curvature learned by P2 (seed 0) is $\kappa_{\text{net}} = 0.418/0.314/0.334/0.314$ (extrapolation) and $0.295/0.224/0.255/0.219$ (interpolation), for $u_x/u_y/\theta/p$; the three-seed means are $0.418/0.321/0.343/0.338$ and $0.286/0.232/0.252/0.265$. The own-data row of Table 12 is the P3 architecture trained with 150 epochs and one seed, the protocol of the public-transfer rows; this table uses 200 epochs and three seeds; the same architecture therefore differs between the tables by 2–6% in extrapolation and 6–12% in interpolation.

split	channel	A	B	P1	P2	P3
extrapolation ($n = 218$)	u_x	0.2289	0.1508	0.0937	0.0867	0.1214
	u_y	0.2733	0.1725	0.1060	0.0951	0.1277
	θ	0.0406	0.0255	0.0141	0.0131	0.0161
	p	0.2689	0.1791	0.1126	0.1007	0.1682
interpolation ($n = 200$)	u_x	0.1530	0.0673	0.0545	0.0507	0.0732
	u_y	0.1796	0.0714	0.0524	0.0488	0.0749
	θ	0.0261	0.0097	0.0078	0.0073	0.0092
	p	0.1792	0.0667	0.0454	0.0410	0.0893

Table 7 reports a U-Net of width 32, 200 epochs, mean of three seeds, identical protocol for all columns (largest standard deviation over every cell: 0.0051).

Three seeds give the resolution. The paired B−P1 difference is positive in every channel in both splits: extrapolation 0.057/0.067/0.011/0.067 and interpola-

tion 0.013/0.019/0.002/0.021 ($u_x/u_y/\theta/p$), and every 95% interval excludes zero (half-widths 0.001–0.020). The FiLM comparison is weaker: in extrapolation the mean differences are 0.006/0.010/0.002/0.012, but the intervals for u_x (−0.001 to 0.013) and u_y (−0.009 to 0.028) include zero; in interpolation the differences are −0.000/−0.003/−0.001/−0.003 and all intervals include zero. We therefore claim a FiLM advantage for the extrapolation mean, not for individual velocity channels and not in interpolation.

Model A's error on the extrapolation split (0.2289) is 9% above the ε-mean reference on the paired set at $\varepsilon = 1.9$ (0.2091, Sec. 5.3); the two are measured on different data sets and with different training sets, but their agreement in magnitude is consistent with a ceiling that is informational rather than architectural. Shallow learners plateau in the same place. A linear ridge regression and an RBF kernel ridge on the raw four-field input—the same information the network gets—reach u_x 0.2282 and 0.2222 in extrapolation, and adding ε as an extra column barely moves them (0.2188 and 0.2171, against 0.0937 for the structured model), because neither can represent a product of an ε-scale with a spatial field. A classical POD-ROM (44–52 modes, 99.9% of the snapshot energy) whose coefficients are regressed on $(\log_{10} Ra, \varepsilon)$ lands at 0.2937 with ε and 0.3001 without, i.e. at the level of doing nothing, and a purely linear surrogate that also sees the POD coefficients of X reaches only 0.2318. The plateau is not capacity—it is that the factor scaling the quantity to be added is absent from the input.

A first-order perturbation proxy. If the correction is linear in ε, the alternative is to solve the linearised compressible problem once per query and scale. That costs one Jacobian assembly and one factorisation of the same discretisation. Measured on the linearised compressible system at $Ra = 10^6$ (45 957 degrees of freedom), one solve takes 1.9 s: 0.12 s assembly, 1.59 s LU factorisation and 0.02 s back-substitution, the remaining 0.17 s being mesh construction and state input/output. Both P and the network are identified offline; the comparison here is per query. The field that solve produces is $P = \lim_{\varepsilon\to 0} D(\varepsilon)/\varepsilon$, the solution of $\mathcal{L}\delta = S_1[X]$. We first estimate P without a new solve: the paired scan already contains $D(\varepsilon)$, so we estimate P by third-order Richardson extrapolation in ε over the solutions at $\varepsilon = 0.4, 0.8, 1.2$. With $\mathcal{P}(\varepsilon) = D(\varepsilon)/\varepsilon$, $P = 3\mathcal{P}(0.4) - 3\mathcal{P}(0.8) + \mathcal{P}(1.2)$, which has $O(\varepsilon^3)$ truncation and the same norm as the $\varepsilon = 0.4$ secant to within 1% in the median over the 20 configurations, in all four channels. The accuracy below is that of this extrapolated field. At 6.3 ms the network is about 300 times cheaper per query than this linear solve, before any allowance for the different hardware. Predicting $\hat{Y} = X + \varepsilon P$ gives 14/15/1.4/15% at $\varepsilon = 1.6$ and 24/27/3.0/28% at $\varepsilon = 1.9$ ($u_x/u_y/\theta/p$), against a do-nothing floor at $\varepsilon = 1.9$ of 39/58/6.0/42%: the linear response removes 34–53% of the gap at $\varepsilon = 1.9$ (computed from the unrounded errors; p gives 34.9%) and leaves the rest to the $O(\varepsilon^2)$ term. The secant at $\varepsilon = 0.4$ predicts 22/26/2.9/26% at $\varepsilon = 1.9$, slightly better than the linear response itself, because the secant already contains part of the quadratic term; it costs one full compressible solve (30.3 s median on this 20-configuration scan; 22.3 s is the 600-sample median of Table 13) and using it as the

linear baseline would flatter that baseline. The network reaches 9.4/10.6/1.4/11.3% on the extrapolation split (Table 7), so the residual left by the linear response is not a small correction. Both approaches replace the same compressible member and therefore save a similar fraction of the 30.3 s nonlinear solve; the first-order solve cannot produce the ε^2 term, while inference costs 6.3 ms against 1.9 s per linear solve.

Direct solution of the $O(\varepsilon)$ system. The extrapolated P above is fitted to the paired data, so it needs a check that is not another fit. Let $F(U;\varepsilon)$ be the steady residual of (1)–(4). One Newton step from the Boussinesq state, $\delta = -J^{-1}F(X)$ with $J = \partial F/\partial U$ evaluated at X, solves the linearised system that defines P: because $F(X) = O(\varepsilon)$, $\delta = D + O(\varepsilon^2)$ and therefore $\delta/\varepsilon = P + O(\varepsilon)$. Measured on the 27 samples of the low-contrast sweep ($\varepsilon = 0.02, 0.05, 0.10$; $Ra = 10^3$–10^5; three independent wall draws each), where the correction is 0.4–5.8% of X, δ/ε reproduces the measured D/ε with median relative errors of 4.7%, 3.8%, 7.2% and 12.2% in u_x, u_y, p and θ (10–90% ranges 0.8–19.6%, 0.5–20.8%, 0.8–39.2%, 1.4–24.6%) and median correlations 0.999, 0.999, 0.998 and 0.994. No extrapolation in ε and no fitted P enter this comparison. The same comparison on the paired scan gives 15% in u_x at $\varepsilon = 0.4$ and 73% at $\varepsilon = 1.9$, rising linearly in ε: a single Newton step is an $O(\varepsilon)$ instrument, and what it leaves behind at high contrast is the $O(\varepsilon^2)$ term. At fixed flow, δ/ε itself moves by 12.5% in u_x between $\varepsilon = 0.1$ and 0.4 and by 153% between 0.4 and 1.9 (median over 17 configurations), while the measured D/ε moves by 9.0% and 102%: the residual and its Jacobian carry ε explicitly through $1/Fr^2 = 1/\varepsilon$, so a solve at finite contrast is not yet the limit and the extrapolation is doing real work. The extrapolated P is then tested against data it never saw. Nineteen of the paired configurations were re-solved at $\varepsilon = 0.1$ and 0.2 from the same random stream, giving 38 solutions that enter no fit; the test holds out the contrast, not the flow. There P predicts the measured D/ε with median relative errors of 2.4%, 1.8% and 3.8% ($u_x/u_y/p$); the θ channel is not predicted (74%), consistent with its unidentifiable exponent. Because $D(0.1)/0.1$ and $D(0.2)/0.2$ differ by only 3.1% in u_x, the three-point combination is estimating a limit that the data approach, not a local secant. Both checks use the same mixed space as the reference solver and one LU factorisation per configuration.

A two-term fit, where it is available. The same low-ε pairs allow a stronger data-driven baseline: fit $\mathcal{P}(\varepsilon) = P + \varepsilon Q$ on $\varepsilon = 0.4, 0.8, 1.2$ for each configuration and predict $X + \varepsilon P + \varepsilon^2 Q$. At $\varepsilon = 1.6$ this gives 3.3/3.9/0.4/3.8% and at $\varepsilon = 1.9$ 11.4/13.9/1.7/13.8% ($u_x/u_y/\theta/p$), against 14/15/1.4/15% and 24/27/3.0/28% for the first-order response. That is the same order as the network's 9.4/10.6/1.4/11.3%, averaged over the extrapolation split; the two are not evaluated on the same samples. The two-term fit is available only when paired low-ε data exist for the target flow. The public-data problem of Sec. 5.7 has no such pairs at the target contrast; there a single-ε data set is all that exists, and the prior—not a quadratic fit—is what makes it reusable.

5.5. The physics prior: $D = \varepsilon P(X)$

Same backbone, only the output structure changes (Table 7 of Sec. 5.4). B keeps the target D and reads ε from a constant input channel; P1 reads ε as well but trains on D/ε and multiplies ε back outside; P3 removes the ε input entirely.

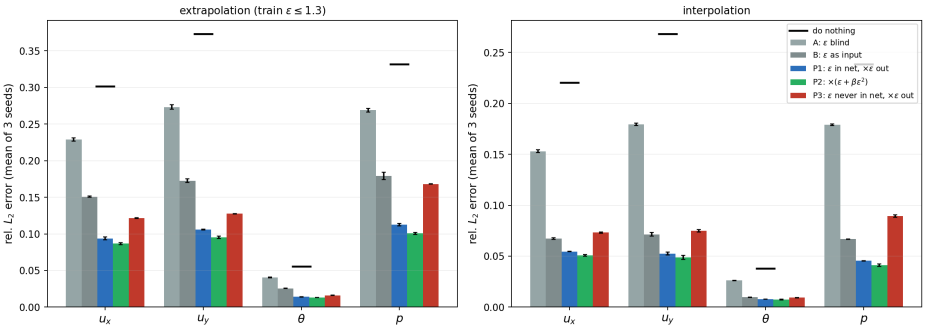


Figure 3: The prior ablation for extrapolation (left) and interpolation (right). A: ε-blind. B: ε as input. P1: ε in the network, multiplied outside. P2: learned $(\varepsilon + \kappa\varepsilon^2)$ factor. P3: ε never enters the network.

Three things happen at once (Table 7, Fig. 3). In extrapolation the factorised output beats a plain network that reads ε from a constant input channel by 1.61× (u_x) and 1.59–1.81× in the other channels, and the quadratic form adds a further 7.1–10.6% on top. That channel is a weak way to condition a network. A FiLM-conditioned U-Net, which modulates every block with ε, reaches 0.1001/0.1155/0.0164/0.1251 in extrapolation and 0.0544/0.0496/0.0073/0.0428 in interpolation, so against that baseline the channel-averaged error of the factorised output is 1.09× lower in extrapolation (1.16× in θ and 1.11× in p in the fixed gauge; the u_x and u_y intervals include zero), while the two are indistinguishable in interpolation. Removing ε from the input as well costs accuracy relative to P1 and FiLM—0.1214 against 0.0937 and 0.1001 in u_x—but still beats the ε-blind network (1.87× channel-averaged on the same 600-sample scan; 1.89× in u_x); what it buys is the property P1 cannot have, a network that can be trained at one ε and applied at another (Sec. 5.9). Finally, the curvature learned by P2 is positive, $\kappa_{\text{net}} = 0.22$–$0.42$, the same sign and order as the ε^2 term of the expansion in Sec. 3—an independent check of the asymptotic expansion.

Practical form. The prior changes only the training target: train any regressor $\mathcal{N}$ on the ratio D/ε instead of on D, and at run time return $\hat{Y} = X + \varepsilon\,\mathcal{N}(X)$ after evaluating the Boussinesq member (or a surrogate for it) to obtain X. For wall heat transfer, add $\lambda \sum |\hat{q} - q| / \sum |q|$ to the loss, where q is the wall-normal gradient of θ; $\lambda = 1$ with the six-point stencil is enough (Sec. 5.8). Supplying ε to $\mathcal{N}$ as well is the most accurate form (P1, 1.09× lower channel-averaged error than a FiLM-conditioned network in extrapolation, with θ and p individually significant and the u_x/u_y intervals including zero). P2 learns a scalar curvature κ_{net} and replaces

ε by $\varepsilon(1+\kappa\varepsilon)$ before the multiplication. Leaving the input out (P3) costs accuracy at fixed data but is what lets one network serve every contrast for which the prefactor is unchanged, because $\mathcal{N}$ then sees only X and re-learns the prefactor from the flow rather than from a tabulated value.

What a single-contrast data set can and cannot identify. Training on D/ε_0 at one contrast does not return P: the target is $P + \varepsilon_0 Q + O(\varepsilon_0^2)$. Applied at a different contrast ε, the leading term carries the error $\varepsilon(\varepsilon_0 - \varepsilon)Q$. Over the paired scan the measured ratio is $|Q|/|P| = 0.56$ in u_x, 0.31 in u_y and 0.79 in p; for $\varepsilon_0 = 1.2$ applied at $\varepsilon = 0.4$ the error in u_x is $0.18|P|$ against a correction of $0.4|P|$ at that contrast. This is the same order as the transfer errors of Sec. 5.7, and it is why the two amendments that help both act on the amplitude: the curvature term $\varepsilon(1+\kappa\varepsilon)$ of P2, and ε as an input (P1), which lets the network absorb the residual ε-dependence. A single-contrast data set cannot separate P from Q by itself; it can only supply $P + \varepsilon_0 Q$.

A Fourier neural operator (1.19 M parameters) shows the same pattern against its ε-blind version: u_x 0.2287 $\rightarrow$ 0.1334 with the prior and ε multiplied back outside the network (1.71×), statistically indistinguishable from an explicit ε input channel (0.1334 ± 0.0004 against 0.1342 ± 0.0008). On u_y and θ the prior is in fact better (0.1387 against 0.1456; 0.0177 against 0.0191); only on p does the ε channel win (0.1536 against 0.1734). The prior's value is therefore not that it beats an ε-aware model—it is that it matches one while needing no ε in the input (Secs. 5.6, 5.9).

5.6. Data efficiency

The learning curve is tabulated in Table S4 of the supplementary material and plotted in Fig. 4.

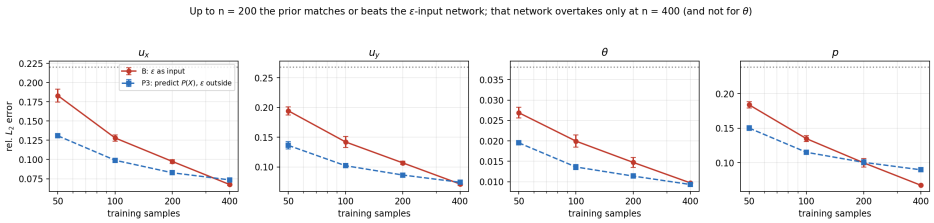


Figure 4: Learning curves. Up to $n = 200$ the prior matches or beats the ε-input network; the latter overtakes only at $n = 400$, and not for θ.

The naive expectation—that a structured model trades in-distribution accuracy for extrapolation—is wrong at small n. At $n = 50$ the prior is 1.4× better than the ε-input network (u_x 0.1308 against 0.1831, a 9σ separation across three seeds), and it is still ahead at $n = 200$ in u_x (7.3σ), u_y (7.6σ) and θ (3.6σ), with p already level (0.13σ) there. The crossover sits between $n = 200$ and 400 in u_x and u_y (log-linear interpolation: $n \approx 315$ and 360, i.e. near 300 in u_x), essentially at $n = 200$ in p, and θ never crosses—at $n = 400$ it is still 2.5σ behind. Concretely, the prior at $n = 200$

reaches what the ε-input network needs ≈ 270 samples (u_x) and 290 (u_y) to reach. So the prior is two things at once: the cheaper model when data is scarce, and—a model that needs no ε in its input, which is what makes a single-ε data set reusable.

5.7. External validation on an independent, publicly released data set

The public release `NeuralConvection_2D_TrainTest` (v2; 4000 training pairs plus three 200-pair held-out splits, CC BY 4.0) was generated by another group, at a single $\varepsilon = 1.2$, $Ra \in [10^2, 3 \cdot 10^6]$ [5].

Convention conversion. Recomputing their published `relL2_*` columns as $\|Y - X\|/\|X\|$ from the tensors reproduces them exactly (0.14233/0.13749/0.01356/0.19070 for sample 0); their pressure amplitude ratio is 0.929 against 0.960 for our fixed pairing, under the same median convention. The comparison below uses $\|Y - X\|/\|Y\|$, the convention of the rest of this paper; the published columns use $\|X\|$ and are reproduced to round-off before conversion.

The law reproduces. Table 8 compares the $\varepsilon = 1.2$ slice of the public release with our own data at the same contrast.

Table 8: External validation at $\varepsilon = 1.2$: public data against our own slice. Medians with p10–p90 in parentheses.

channel	public 4000 (median, p10–p90)	our data at $\varepsilon \in [1.1, 1.3]$ ($n = 64$)
u_x	0.1797 (0.112–0.300)	0.1839 (0.126–0.284)
u_y	0.2435 (0.141–0.403)	0.2250 (0.150–0.395)
θ	0.0228 (0.001–0.071)	0.0232 (0.001–0.056)
p	0.1858 (0.117–0.246)	0.1889 (0.128–0.262)

Even the secondary trends reproduce. The correlation of R with $\log_{10} Ra$ is +0.133, −0.419, +0.647 and −0.198 on their $\varepsilon \equiv 1.2$ data and +0.032, −0.511, +0.662 and −0.263 on our $\varepsilon \in [1.1, 1.3]$ slice (64 samples, $u_x/u_y/\theta/p$). The θ channel—the one whose scatter we explain next—has $r = +0.66$ on ours and +0.65 on theirs, from independently generated data.

Five splits, four prefactors, one predictable pattern. The release publishes three held-out splits generated by the same generator and the same solver commit (`v6_freefall_20260921`, `raw_both_deramp` pressure convention; the OOD split carries the later tag `v6_freefall_ood_20260925`): `test_in_distribution` (200 random-wall pairs), `test_ood` (200 random-wall pairs whose vertical-wall Fourier dials lie outside the training range) and `test_legacy` (200 pairs of the classical cavity: constant hot and cold side walls, adiabatic top and bottom). The release also contains `train_legacy`, 5000 classical-cavity pairs, which we recompute below. Recomputing all of `relL2_u_x`, `relL2_u_y`, `relL2_T`, `relL2_p` for every sample

of all five splits (38 400 values: 9600 samples × 4 channels) under our conventions reproduces the published numbers to within 1.2×10^{-7} (float32 round-off), so nothing below is a convention artefact. At the fixed $\varepsilon = 1.2$ the medians of $R = \|Y - X\|/\|Y\|$ are listed in Table S5 of the supplementary material for all five splits, and in Table 8 for the two that matter here.

The exponent cannot be re-tested here—every split sits at one ε—but the prefactor is put under pressure and it moves: by 1.66× (u_x), 1.38× (u_y), 4.05× (θ) and 1.07× (p) between the classical cavity and the random-wall families at the same ε, and `train_legacy` and `test_legacy` agree to three digits in u_x and p (θ: 0.0926 vs 0.0836, 10%), which is the internal consistency check for the whole comparison. That is a statement about the flow, not a defect of the law, and the θ channel makes the mechanism explicit. Writing $\hat{u} = \max \|\boldsymbol{u}_X\|$ for the peak speed of the Boussinesq field, which is available to any surrogate, the θ-channel prefactor collapses on $\hat{u}$ (Fig. 5); the fits are in Table S6 of the supplementary material.

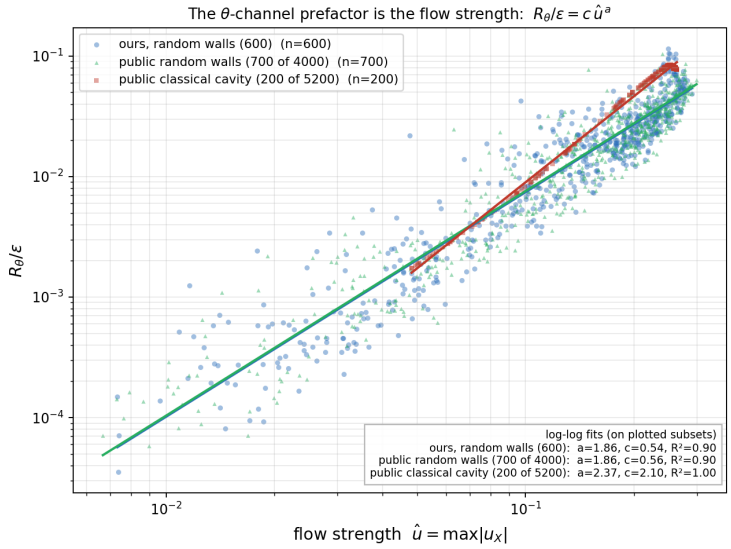


Figure 5: The θ-channel prefactor collapses on the peak Boussinesq speed $\hat{u}$ across two independent generators and two boundary-condition families.

Two independent groups, two independent generators, six data sets: the random-wall families all give $c = 0.54$–0.69, the classical cavity—whose flow is a single strong circulation with $\hat{u}$ 41% above the public random-wall median and 56% above ours—gives $c = 2.06$–2.10 . The same reading explains the residual scatter in $u_x/u_y/p$ in Sec. 5.1: A is a functional of the flow, and the flow is in the input.

5.8. What the correction does to the wall heat transfer

Every error quoted so far is a field error. For a heat-transfer application the quantity that matters is the Nusselt number, and the public classical cavity answers

the question directly at $\varepsilon = 1.2$, because the two members share mesh and boundary conditions. We compute $Nu = -\langle \partial\theta/\partial n \rangle$ at the hot wall with a six-point one-sided stencil. On the Boussinesq member the same flux can be obtained without any wall stencil, from the horizontally integrated flux $\int [u_x\theta - (Ra\,Pr)^{-1/2}\partial_x\theta]\,dy$, which is independent of x to 0.1% at the three interior planes $x = 0.25, 0.5, 0.75$; the wall estimate agrees with it to 0.35%, which indicates the discretisation error of the estimator; the integral change reported below is 1.5 times that error.

Table 9: Wall heat transfer at $\varepsilon = 1.2$ over the 200 classical-cavity samples of the public data set. The local change is $\Delta Nu(y)/Nu(y)$ along the hot wall. The integral change is the mean of the per-sample ratios at the six-point stencil, with the median and the 10–90% range; the ratio of the ensemble means is +0.67%. The stencil range is the two- to eight-point one-sided estimate of the wall gradient.

quantity	value	spread over samples
Nu, Boussinesq X	3.514	—
Nu, compressible Y	3.538	—
integral change $\Delta Nu/Nu$	+0.54%	median +0.58%, p10–p90 +0.06 to +0.92%; per-sample mean +0.35 to +0.62% over stencils
local change, L_1 norm over the wall	4.25%	p90 7.33%; unchanged for two to eight points
local change, pointwise	−0.65% (median)	p10–p90 −9.1 to +9.8%

Table 10: Wall heat transfer of the public-data surrogate at $\varepsilon = 1.2$ (200 classical-cavity samples). The surrogate is trained on the 4000 public training pairs and evaluated without retraining. T error is relative L_2; Nu error is the mean absolute error of the integrated wall Nusselt number divided by the mean true Nu; local flux is the mean absolute error of the wall-normal flux profile divided by its mean absolute value. For doing nothing the absolute error equals the mean signed change, because all 200 per-sample changes are positive. The last row adds a relative L_1 penalty on the wall-normal gradient to the training loss (weight 1, two seeds).

model	T error	Nu error (hot)	Nu error (cold)	local flux (hot/cold)
do nothing ($\hat{Y} = X$)	7.2%	0.67%	0.64%	6.0/5.2%
plain D	0.10%	3.61%	2.42%	4.4/3.7%
prior $D = \varepsilon P(X)$	0.10%	2.56%	1.72%	4.3/3.6%
prior + wall-flux loss	0.13%	0.11%	0.10%	1.0/0.9%

FiLM is not a separate row here: the public data have a single ε, so its conditioning input is constant and adds no information beyond the plain model. The FiLM comparison across ε is the field-error comparison of Table 7.

The two statements are different in kind. The integral heat transfer is insensitive: the compressible member carries more heat in all 200 samples, by 0.01–0.94% (median 0.58%, 10–90% range 0.06–0.92%). The +0.54% of Table 9 is the mean of the per-sample ratios at the six-point stencil, and the ratio of the ensemble means is

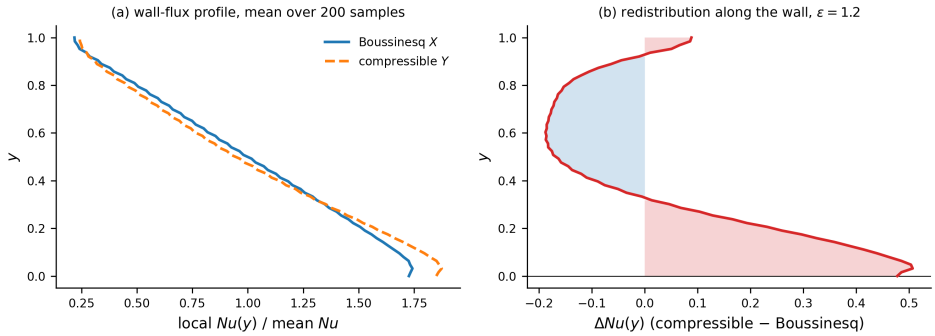


Figure 6: Wall heat transfer of the pair at $\varepsilon = 1.2$. (a) Local Nusselt number along the hot wall, normalised by its mean, for the two members. (b) Redistribution along the wall: the compressible member carries more heat in the lower half and less in the upper half, while the integral changes by less than 1%.

+0.67%. Across two- to eight-point stencils the per-sample mean spans +0.35 to +0.62% and the ratio of the ensemble means +0.30 to +0.90%. The local wall flux is not: the correction redistributes it by 4.25% in L_1 (p90 7.33%), with pointwise excursions of ±10%, and that number is the same for every stencil order. On average the correction adds heat over the lower half of the cavity and removes it over the upper half (Fig. 6b); the integral is the Nu-weighted mean of this redistribution, which is why it cancels. Doing nothing predicts the integral heat load to 0.7% only because it predicts no change at all; it is not a heat-transfer surrogate. Table 10 evaluates the public-data prior directly. The prior reduces the temperature-field error from 7.2% to 0.10% in relative L_2, and the local wall-flux L_1 error from 6.0% to 4.3% (hot wall) and from 5.2% to 3.6% (cold wall), but its integrated wall Nu error is 2.6%, larger than the 0.7% of doing nothing: the wall-normal derivative carries a bias that the field norm does not see. Adding a relative L_1 penalty on the wall-normal gradient of the predicted θ to the training loss removes it. With unit weight and the same six-point stencil, the integrated wall-Nu error falls to 0.11% and the local wall-flux error to 1.0%, at a cost of 0.10% → 0.13% in the field norm; two seeds give 0.10–0.11%. The gain is not tied to the training stencil: evaluated with four- and eight-point stencils, the same models give 0.3–0.9%, against 1.8–3.6% for the field-only prior and 0.4–0.9% for doing nothing. The size of the gain depends on the stencil, its sign does not. The field statistic behind the cancellation is large: on the same 200 samples the temperature field differs by 7.2% in global relative L_2 (8.4% median per sample), so a 7–8% change in the field produces a 0.6% change in its wall-normal derivative.

The relative gap is not a meaningful statistic for every configuration. Of the 20 paired configurations, eight have $|Nu_X| \leq 1$ at $\varepsilon = 1.9$ and nine have negative Nu_X; there the sign or the small magnitude of the denominator, not the correction, sets $\Delta Nu/Nu$, and the all-configuration mean +5.06% is not a physical heat-transfer result. The five isothermal-sidewall configurations have $Nu_X = O(1)$ at every

contrast. This family is the classical cavity, fixed before the analysis; it is not selected by the outcome. On those, the mean integral change is +0.51% (sd 0.37%) at $\varepsilon = 1.2$ and −1.89% (sd 1.94%) at $\varepsilon = 1.9$. The point estimate changes sign, but with five configurations the 95% interval of the $\varepsilon = 1.9$ mean is [−4.3, +0.5]% and includes zero; we report the sign change as a descriptive trend consistent with the ε^2 term, not as a significant effect. A Boussinesq-trained surrogate cannot produce it because it has no access to the compressible correction.

The Rayleigh-number dependence of the integral correction. The five Rayleigh numbers span +0.01 to +0.87% at $\varepsilon = 1.2$ and −5.1 to −0.01% at $\varepsilon = 1.9$ (Table 11). At $Ra = 10^2$ the change is at or below 0.01%; at $Ra = 10^3$ it stays within 1%; at $Ra \geq 10^4$ it reaches −5%. The 15 random-wall configurations are excluded: eight have $|Nu_X| \leq 1$ at $\varepsilon = 1.9$, where the sign or the small magnitude of the denominator, not the correction, sets the ratio. A model calibrated on Nu alone therefore does not transfer across Ra at fixed ε; the local field is the quantity that carries the correction.

Table 11: Integral Nusselt change $\Delta Nu/Nu$ (percent) for the five isothermal-sidewall configurations, one at each Ra. These are the only configurations with a positive $O(1)$ denominator at every contrast; this is the classical-cavity family, fixed before the analysis. Random-wall families are excluded because 8 of 15 have $|Nu_X| \leq 1$ at $\varepsilon = 1.9$. At $Ra = 10^2$ the entries are below the 0.35% estimator error and are listed only for completeness. At $Ra = 10^6$ the $\varepsilon = 1.2$ entry falls to +0.44% on a 256^2 mesh and the $\varepsilon = 1.9$ entry to −2.10% on a 192^2 mesh.

Ra	$\varepsilon = 0.4$	0.8	1.2	1.6	1.9
10^2	−0.00	+0.00	+0.01	+0.01	−0.01
10^3	+0.07	+0.28	+0.59	+0.76	−0.98
10^4	+0.07	+0.20	+0.27	−0.38	−5.14
10^5	+0.16	+0.48	+0.87	+0.91	−1.81
10^6	+0.24	+0.51	+0.82	+0.83	−1.50

5.9. Cross-data-set transfer: the payoff, and its boundary

Two public single-ε data sets are available (Table 12, Fig. 7): 4000 random-wall pairs (our boundary-condition family) and 5000 classical-cavity pairs (a different one). Both sit at $\varepsilon \equiv 1.2$. We train the same U-Net on each—once with the plain output (target D), once with the prior (target D/ε, and ε multiplied back outside the network)—and apply the result to our own 600 samples with no retraining, at ε from 0.2 to 1.9, values the training data never contained.

Three readings, in order of how much they matter for practice.

- Within the family, the prior is what makes a single-ε data set reusable. The identical architecture, budget and data give 0.1443 (u_x) with the plain output and 0.0932 with the prior—a factor 1.55—and 3.2× better than doing nothing,

Table 12: Transfer from a single-ε public data set to our own $\varepsilon \in [0.2, 1.9]$ samples, on the extrapolation and interpolation test splits. Entries are $u_x/u_y/\theta/p$. Both public sets are at $\varepsilon \equiv 1.2$. The "our own training split" row uses 382 training samples for the extrapolation column and 400 for the interpolation one, with a prior that has no ε input (150 epochs, one seed), the same protocol as the public rows above; the P1/P3 columns of Table 7 use 200 epochs and three seeds.

trained on	model	extrapolation ($n = 218$)	interpolation ($n = 200$)
do nothing	$\hat{Y} = X$	0.3018/0.3731/0.0556/0.3320	0.2203/0.2683/0.0381/0.2386
public 4000, random walls	plain	0.1443/0.1730/0.0226/0.1838	0.1199/0.1441/0.0189/0.1435
public 4000, random walls	prior	0.0932/0.1085/0.0119/0.1322	0.0657/0.0769/0.0077/0.0891
our own training split	prior	0.1250/0.1321/0.0170/0.1719	0.0791/0.0811/0.0103/0.0947
public 5000, classical cavity	plain	0.2919/0.3711/0.0490/0.3192	0.2444/0.3612/0.0484/0.3217
public 5000, classical cavity	prior	0.3092/0.4730/0.0561/0.3280	0.2281/0.3498/0.0422/0.2584

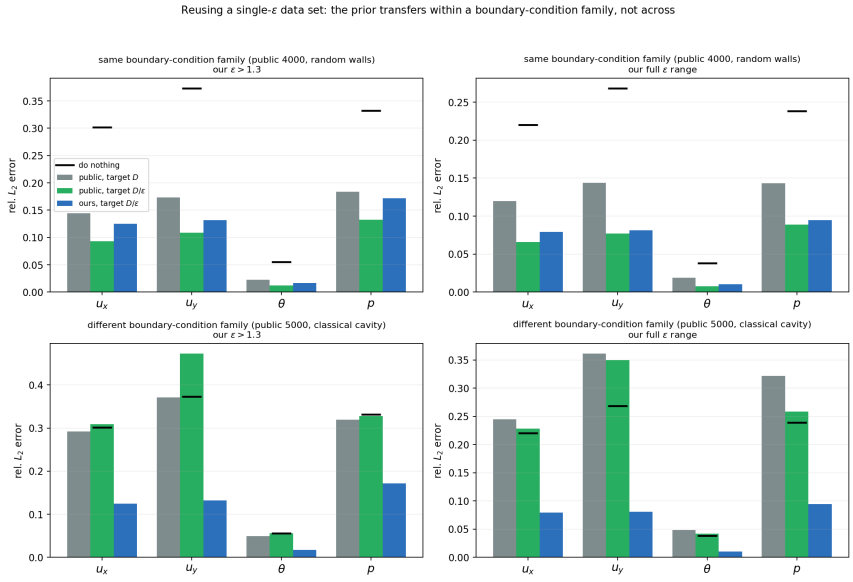


Figure 7: Reusing a single-ε data set. Within the boundary-condition family the prior transfers and beats the plain model; across families it fails, and can be worse than doing nothing.

on ε the data never contained. It is even better than the same prior trained on our own 382 samples (0.1250), because it sees ten times more configurations. This is the concrete content of "the ε dependence need not be in the data".

- The prior also fixes the failure mode of the plain model. The plain model's error at unseen ε is largest where the training ε is furthest away (its correction is a fixed field, Sec. 5.3), whereas the prior's error stays proportional to how wrong P is, uniformly in ε.

- Across boundary-condition families the prior does not rescue transfer. Trained on the classical cavity, the plain model lands at 0.2919 against a floor of 0.3018—it learned little that transfers—and in extrapolation the prior is worse than the plain model in all four channels (0.3092/0.4730/0.0561/0.3280 against 0.2919/0.3711/0.0490/0.3192). In interpolation the prior does beat the plain model in all four channels (0.2281/0.3498/0.0422/0.2584 against 0.2444/0.3612/0.0484/0.3217), but it still does not beat doing nothing. The failure is the one Sec. 5.7 predicts: A is a functional of the flow, and a surrogate trained on one family does not know another family's A. The classical cavity's peak Boussinesq speed is 41% above the random-wall median, so $\hat{u}$ separates the two families; it does not, however, order the per-sample error within the public random-wall splits under the per-channel relative L_2 metric used here (Spearman −0.20; partial correlation 0.09 at fixed Ra). We therefore use $\hat{u}$ as a family-level diagnostic, not as a within-family transfer law.

5.10. A second geometry (aspect ratio 2:1, 104 samples)

Same generator, same Ra and ε ranges, cavity twice as wide (Fig. 8). The ε exponent is consistent within its uncertainty in the ε-only fit ($m = 1.09/1.16/0.93/1.08$ for $u_x/u_y/\theta/p$, 0.7, 1.8, 0.3 and 0.7 standard errors away from the unit-square values; these are combined standard errors of the two fits, with the 104-sample geometry fit contributing 0.100, 0.087, 0.268, 0.099 and the 627-sample baseline 0.018, 0.019, 0.077, 0.015; the refitted values themselves move by 7–17%; the two-parameter values are in Table S2), while the prefactor moves ($A = 0.175/0.175/0.008/0.222$ versus the unit-square values of Table 4, 0.147/0.193/0.010/0.148). The Rayleigh trend is stronger for u_x and p here ($q = -0.14, -0.15$ vs 0.00, −0.03) and reproduces for u_y and θ. For u_x, u_y and p the exponent is consistent across geometry within its uncertainty (p in the fixed gauge); the θ slope is also consistent in this comparison but remains subset-dependent (Sec. 5.13). Calibration of A is not—which is exactly why the surrogate takes X as input rather than a table of A. Changing the fluid instead (Pr 0.71 → 4.38) leaves the paired exponent within 1.5% and the paired median R within 5%, and the fitted prefactors move by at most 3.3% except p, which moves by 6.1% (Sec. 5.13).

5.11. Solver noise is not the source of the law

- Convergence quality over 600 samples (compressible member): median relative change 3.75×10^{-6}, p90 1.09×10^{-4}, worst 7.04×10^{-4}.

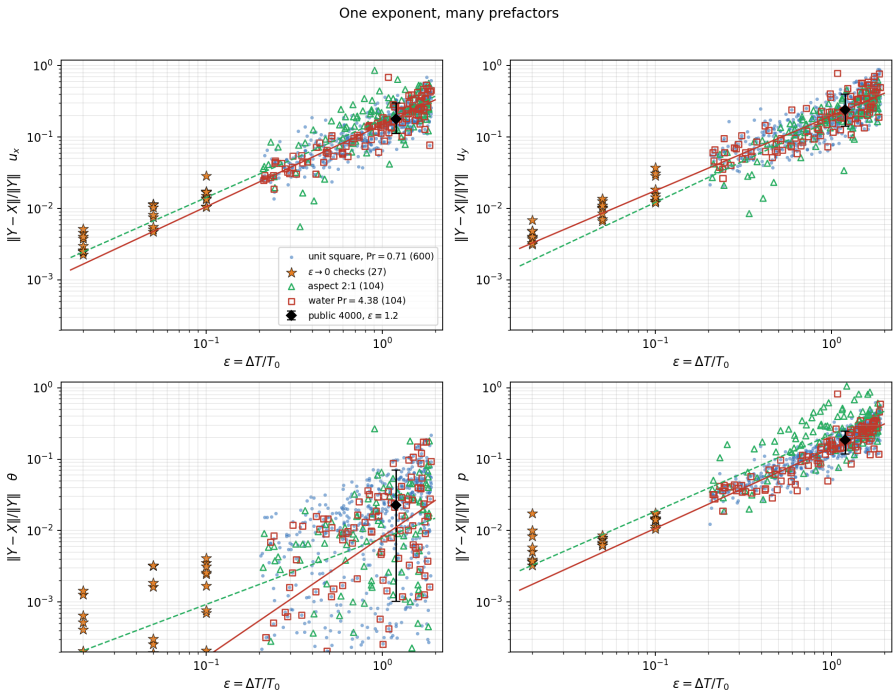


Figure 8: One exponent, many prefactors: unit square ($Pr = 0.71$), aspect 2:1, $Pr = 4.38$, and the public data set at $\varepsilon = 1.2$.

- Dropping the 65 samples with relative change $> 10^{-4}$ changes the linear u_x fit from $0.1809\varepsilon - 0.0212$ to $0.1813\varepsilon - 0.0221$—unchanged. The negative intercept is a property of this linear fit; the power-law fit of Table 4 has an intercept consistent with zero.

- Re-running the 10 worst samples with `steady_tol` tightened from 10^{-6} to 10^{-8} leaves the solution bit-identical ($|\Delta Y|/|Y| = 0$). The reason is structural: the compressible march terminates on the zero-iteration guard, not on `steady_tol`, so the residual reported is the fixed-point residual of the discrete solution, not unfinished iteration.

5.12. Cost and speed-up

Table 13 gives the wall-clock cost. A paired solve splits into a Boussinesq member (median 11.7 s) and a compressible member (22.3 s); the surrogate replaces only the second. Per pair the cost therefore falls from 33.9 s to 11.7 s, a factor 2.9, and the comparison of one inference with one compressible solve (22.3 s/6.3 ms = 3.5×10^3) describes only that member. The amortised regime depends on how many flows are swept, because X does not depend on ε. For a single flow and $N = 100$ contrasts, one Boussinesq solve plus 100 inferences costs $16.0 + 0.63 = 16.6$ s against $16.0 + 100 \times 37.5 = 3.77 \times 10^3$ s, a factor 2.3×10^2. The scan measured

Table 13: Wall-clock cost, 64^2 mesh. Solver rows use eight CPU threads; the surrogate runs on an RTX 4060 Laptop GPU (batch 1) or one CPU thread. The two members are timed separately over the 600 samples; the rows below them are surrogate timings or totals derived from the measured medians. The 100-sample scan rows use that scan's own timings: the component medians are 16.0 s (Boussinesq) and 37.5 s (compressible), while the median of their per-pair sum is 55.8 s; medians are not additive.

quantity	median	mean	p90	max
paired solve, 600 samples	33.9 s	38.0 s	59.9 s	510 s
Boussinesq member alone	11.7 s	13.1 s	21.7 s	—
compressible member alone	22.3 s	25.0 s	37.3 s	—
paired solve, 100-sample ε scan (per sample)	55.8 s	67.7 s	136 s	202 s
surrogate, GPU, batch 1	6.3 ms			
surrogate, GPU, batch 256	0.49 ms/sample			
surrogate, CPU, 1 thread	14.0 ms			
end-to-end pair (Boussinesq + surrogate)	11.7 s			
one flow, 100 contrasts (scan timing)	16.6 s			
20-flow × 5-contrast scan, surrogate total	321 s			

here is 20 flows × 5 contrasts and was run in a separate session, 1.7 times slower per sample; its own timings are 16.0 s for the Boussinesq member and 37.5 s for the compressible one; the median of their per-pair sum is 55.8 s. On the same-reuse basis used above, the baseline is $20 \times 16.0 + 100 \times 37.5 = 4.07 \times 10^3$ s and the surrogate needs $20 \times 16.0 + 100 \times 0.0063 = 3.21 \times 10^2$ s, a factor 13. If the baseline is instead the scan as actually run, which repeats the Boussinesq solve at every contrast, $100 \times 55.8 = 5.58 \times 10^3$ s, the ratio is 17; we quote the conservative same-reuse 13× in the abstract. Generating the training split (about 400 paired solves, 3.8 h at the measured median) is repaid after roughly four 20-flow × 5-contrast scans, each of which saves 3.75×10^3 s. The solver runs on eight CPU threads and the surrogate on a GPU; the two are not the same hardware.

5.13. A third Prandtl number: $Pr = 4.38$

Pr and ε do not mix. Pr enters the coefficients of both members ($Re^{-1} = (Pr/Ra)^{1/2}$) and hence the Boussinesq solution X, but the ε-expansion of Sec. 3 never multiplies the two. The prediction is therefore sharp and asymmetric: the exponent must not move, while the prefactor may, because X itself has changed.

We generated 104 further pairs at $Pr = 4.38$, the Prandtl number of water near 40°C. The generator changes Pr at fixed Ra in the dimensionless formulation, so $Re = (Ra/Pr)^{1/2}$ and the thermal diffusivity change with it; the equation of state and γ are those of the air run. The study isolates the Pr axis and does not claim water properties (Table 14, Fig. 9). The pairs are bit-identical to the first 104 samples of the main batch in every other respect: same seed stream, hence the same wall-temperature Fourier draws, the same Ra and the same ε. The two families are therefore a strict paired experiment in which Pr is the only changed parameter.

Table 14: The fluid axis: 104 strictly paired configurations at $Pr = 0.71$ and $Pr = 4.38$ (same seed stream, same Ra and ε). Both columns are fitted on the same 104 configurations over their full $\varepsilon \in [0.21, 1.89]$ range; the exponents are steeper than the full-window values of Table 4 because the sample subset differs, and the two are not directly comparable. The exponent agrees between the two fluids within 1.5%, the paired median ratio of R is 0.95–1.01, and the fitted prefactors move by 3.3% (u_x), 0.2% (u_y), 1.2% (θ) and 6.1% (p).

channel	A, $Pr = 0.71$	A, $Pr = 4.38$	m, $Pr = 0.71$	m, $Pr = 4.38$
u_x	0.1448	0.1496	1.157	1.150
u_y	0.1981	0.1985	1.047	1.049
θ	0.0081	0.0082	1.668	1.686
p	0.1519	0.1426	1.109	1.124

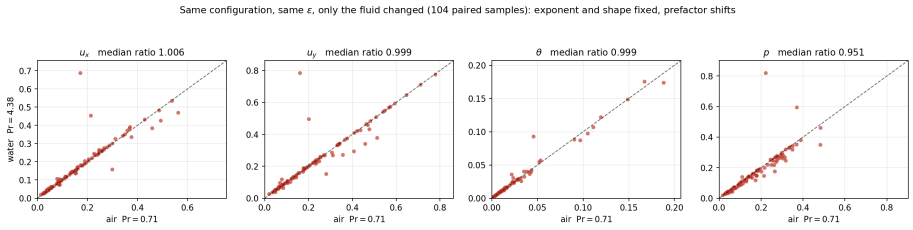


Figure 9: Same configuration, same ε, only the Prandtl number changed: $Pr = 0.71$ against $Pr = 4.38$, 104 paired samples. The panel labels follow the original run names; both runs use the same equation of state and the same dimensionless transport model; only Pr (and hence the thermal diffusivity at fixed Ra) differs.

On the 104 matched configurations the paired ratios $R(Pr = 4.38)/R(Pr = 0.71)$ have medians 1.006, 0.999, 0.999 and 0.951 ($u_x/u_y/\theta/p$; p10–p90 0.99–1.12, 0.90–1.03, 0.95–1.06, 0.82–1.00), and the paired exponents agree within 1.5%: 1.150 vs 1.157, 1.049 vs 1.047, 1.686 vs 1.668, 1.124 vs 1.109. A 6.2-fold change in Prandtl number leaves the discrepancy of the same configuration unchanged to 0.1% in θ, 0.6% in u_x, 0.1% in u_y and 5% in p.

Pr is therefore not a break-symmetry axis either, and the law's two ingredients separate cleanly: ε sets the exponent, the flow sets the prefactor, and the fluid changes neither by more than 3.3% for u_x, u_y and θ and 6.1% for p on the paired comparison. Keeping the strict pairing matters: the matched fits give θ 0.0081 → 0.0082 and p 0.1519 → 0.1426, while u_x and u_y move by 3% and 0.2%. This is a much smaller fluid effect than the geometry (Sec. 5.10) or the boundary-condition family (Sec. 5.7), which move A by up to 4×.

5.14. A three-dimensional check

The law is calibrated in two dimensions. As a preliminary check of whether the exponent survives in three, we generated fifteen paired steady solutions with the same generator: three fixed flows—a three-cell Voronoi wall pair, a two-cell wall pair and a constant-wall cavity—each solved at $\varepsilon = 0.2$, 0.4, 0.8, 1.2 and 1.6,

with $Ra = 10^4$, $Pr = 0.71$ and $res = 16$ (33^3 velocity nodes, 17^3 temperature and pressure nodes). The pairing is exact: within each flow the wall temperatures are the same seed stream scaled affinely about T_0, so the low-fidelity member is the same solution up to the ε-blind rescaling of Sec. 2.3. For the pressure channel we use the same fixed gauge as in two dimensions: subtract the mean, remove a linear fit in y (the gravity direction), and flip the sign; the horizontal z direction is not deramped. The p exponents below are therefore fixed-gauge numbers, as in Table 4.

Table 15 and Fig. 10 give the result. In every channel and every flow a power law fits the five ε values in log–log coordinates with $R^2 > 0.995$, and the per-flow exponents are 1.091 ± 0.014 (u_x), 1.052 ± 0.021 (u_y), 1.130 ± 0.015 (θ) and 1.141 ± 0.017 (p)—within 0.16 of unity. Against the full-window 2-D fit of Table 4 the flow means differ by 0.07 (u_x), 0.06 (u_y) and 0.13 (p); against a 2-D fit restricted to the same $\varepsilon = 0.2$–1.6 window ($1.04/0.95/1.20/1.08$ for $u_x/u_y/\theta/p$, 485 samples) the differences are $0.05/0.11/0.07/0.06$. The second comparison is noisier because a narrow-window regression there also measures the between-flow prefactor scatter (up to 4×, Sec. 5.7). Both are consistency checks, not a converged validation. The prefactor is not universal here either: across the three flows it varies by 3.8× (u_x), 2.1× (u_y), 11× (θ) and 2.0× (p), which is the same statement as Sec. 5.7, that A is a functional of the flow. What transfers to three dimensions is the exponent, which is what the factorisation predicts. With fifteen samples at a single Ra and a single resolution, this is a consistency check, not a converged three-dimensional validation.

A POD-ROM check of the prior. The same fifteen samples also support a linear surrogate check. We train a POD-ROM with at most four modes on three flows and four contrasts (twelve samples) and test on the held-out contrast (three samples), repeating for each of the five contrasts. The plain ROM predicts D from $[1, \varepsilon, \boldsymbol{c}_X]$, where $\boldsymbol{c}_X$ are at most four POD coefficients of the Boussinesq field; the prior ROM predicts $P = D/\varepsilon$ from $[1, \boldsymbol{c}_X]$ and multiplies by ε outside. At most four modes are used, fewer than the twelve training samples. Averaged over the five held-out contrasts, the prior lowers the relative error in every channel (Table 16): u_x from 0.168 to 0.081, u_y from 0.190 to 0.150, θ from 0.034 to 0.027 and p from 0.163 to 0.098. The gain is largest at $\varepsilon = 0.2$ and 0.4 (14× and 5× in u_x) and is not uniform: at $\varepsilon = 1.2$ the ε-aware plain ROM is better in all four channels. We read this as a check that the factorisation remains useful in three dimensions at small contrast, not as a claim of uniform superiority.

The check is limited to $Ra = 10^4$, $res = 16$ and $\varepsilon \leq 1.6$. The fifteen pairs cost 12–32 min each (median 16 min) of outer wall-clock time on this machine. We therefore report the exponent transfer, not a three-dimensional sweep.

6. Limitations

1. Two dimensions for the main sweep. The law is calibrated on 2-D solutions; a 2× mesh refinement moves the exponent by less than 0.005 at $Ra = 10^4$ and by at most 0.015 at $Ra = 10^6$, where the largest pointwise change in R is

Table 15: Three-dimensional check: fifteen strictly paired samples, three fixed flows × five ε at $Ra = 10^4$, $Pr = 0.71$, $res = 16$. Per-flow exponents are power-law fits to the five ε values of that flow. Flows are ordered three-cell Voronoi, two-cell Voronoi, constant wall. The prefactor is listed per flow to show that it does not transfer between flows.

channel	m (mean±s.d. over flows)	per-flow m	A per flow
u_x	1.091 ± 0.014	1.108/1.083/1.083	0.068/0.186/0.260
u_y	1.052 ± 0.021	1.047/1.034/1.075	0.123/0.160/0.255
θ	1.130 ± 0.015	1.126/1.147/1.118	0.006/0.025/0.066
p	1.141 ± 0.017	1.121/1.152/1.150	0.096/0.185/0.195

Table 16: Three-dimensional POD-ROM check on the fifteen paired samples. Leave-one-contrast-out: train on three flows × four ε (twelve samples), test on the held-out ε (three samples), repeated for the five contrasts. Entries are mean relative L_2 errors over the five folds (mean ± s.d.); the ROM uses at most four POD modes and ridge regression ($\lambda = 10^{-4}$). The gain is the plain-ROM error divided by the prior-ROM error.

channel	do nothing	plain ROM	prior ROM	gain
u_x	0.1934±0.132	0.1676±0.071	0.0807±0.061	2.08
u_y	0.1880±0.127	0.1901±0.046	0.1497±0.100	1.27
θ	0.0362±0.025	0.0338±0.008	0.0270±0.019	1.25
p	0.1617±0.117	0.1634±0.065	0.0980±0.077	1.67

3.3% (Sec. 5.1). The preliminary 15-sample 3-D check of Sec. 5.14 agrees with the 2-D exponent within 0.16 over the same $\varepsilon = 0.2$–1.6 window, but covers one Rayleigh number, $res = 16$ and $\varepsilon \leq 1.6$; the POD-ROM baseline is a linear model with at most four modes, not a deep operator. Each 3-D pair costs 12–32 min of outer wall-clock time (median 16) on this machine. A 3-D sweep at higher Ra or resolution remains out of reach here.

2. Ma is inert here, so one parameter is free rather than two. At larger Ma this would have to be redone.
3. $\varepsilon < 2$, imposed by the equation of state (the cold wall must stay positive).
4. The power-law fits of Sec. 5.1 leave 16/18/78% of the variance unexplained in $u_x/u_y/\theta$ and 13% in p; the χ form of Sec. 5.1 raises R^2 from 0.19 to 0.78 and the $\hat{u}$ form of Sec. 5.7 to 0.90, and the same residual flow-dependence remains in the other three channels.
5. θ is not part of the common-exponent claim. Its free power-law fit is not identifiable at this sample size: $R^2 = 0.22$, slope 1.024 over the full window and 1.67 on the 104-configuration subset, and the fitted amplitude is not predictive (0.012 at $\varepsilon = 1.2$ against a public median of 0.0228). We make no exponent claim for θ.
6. The pressure channel is quoted in the fixed gauge and hydrostatic correction of Sec. 5.2. Its exponent is not gauge-invariant, and a different post-processing gauge would change it.

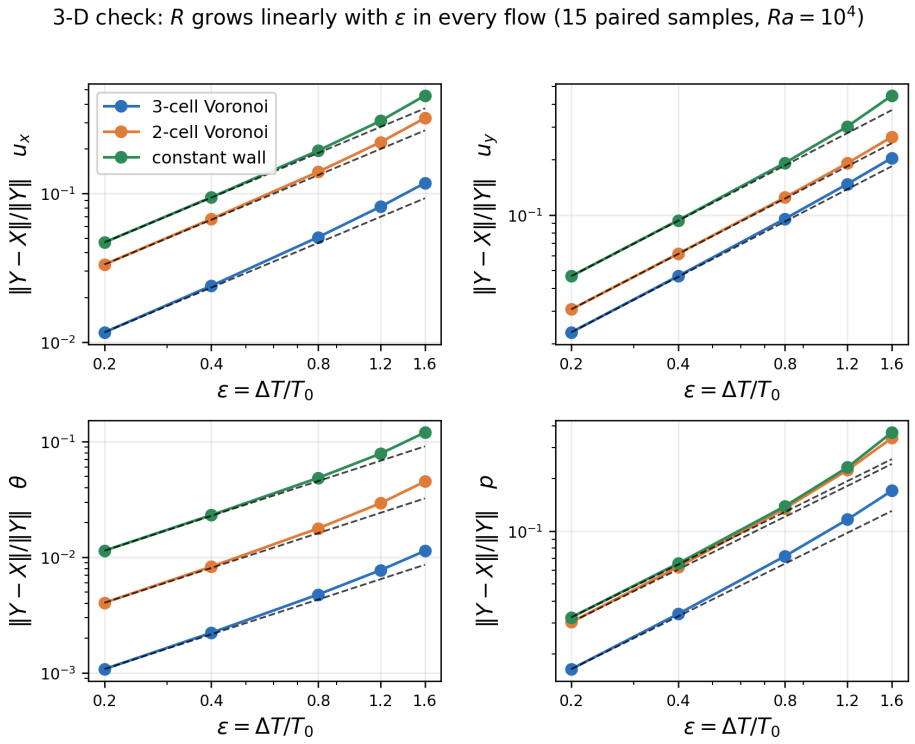


Figure 10: The three-dimensional check in log–log coordinates: fifteen paired samples, three fixed flows and five temperature contrasts. Dashed lines are a slope-one reference anchored at the lowest ε of each flow.

7. The prefactor A is configuration-dependent and must be learned: it moves with the geometry (Sec. 5.10) and by 1.66× (u_x) / 1.38× (u_y) / 4.05× (θ) / 1.07× (p) between boundary-condition families of an independent data set (Sec. 5.7), while it is insensitive to the fluid (Pr, Sec. 5.13).
8. The field norm does not certify the wall-normal derivative. A prior trained on the field alone reaches 0.10% in relative L_2 but leaves a 2.6% integrated wall-Nu error at $\varepsilon = 1.2$, worse than the 0.7% of doing nothing. The wall-flux penalty in Sec. 5.8 removes it (0.11%), but the loss has to be stated; a field-norm metric alone does not detect the failure.
9. The earlier public release used a pinned $Fr = 1.095$, which puts its $Ra_{\text{eff}} = Ra\,Fr^2/\varepsilon$ 0.08% below ours at $\varepsilon = 1.2$ (1.2/1.199 = 1.0008); the v2 release corrects this and the two now agree.

7. Conclusion

We set out to build a surrogate for compressible natural convection and found that the object it has to reproduce—the gap between the Boussinesq solution and

the compressible one—factorises: ε sets its leading-order scaling, and the flow sets its amplitude and spatial shape.

1. One break-symmetry parameter. In this non-dimensionalisation $Fr^2 = \Delta T/T_0$ identically, so the break-symmetry axis is $\varepsilon = \Delta T/T_0$; Mach-number effects are $O(Ma^2)$ and are below the resolution of the sampled range. This does not make the amplitude universal: $P(X)$ carries the flow dependence.
2. A leading-order law. $R = \|Y - X\|/\|Y\| = A\varepsilon^m$ with $m = 1.023 \pm 0.018$ and 0.995 ± 0.019 (u_x, u_y), an intercept indistinguishable from zero, the exponent quoted over the full two-decade window (from $\varepsilon = 0.8$ upwards the same points give 1.23–1.32, the ε^2 curvature of Sec. 5.1), and a 134-family boundary-condition description (133 dummies) that adds only 2–3 pp of adjusted R^2 to the velocity and pressure channels. The fit leaves 16/18/78/13% of the variance unexplained in $u_x/u_y/\theta/p$; θ collapses on χ rather than on ε alone. The pressure channel gives $m = 1.009 \pm 0.015$ in the fixed gauge of Sec. 5.2; this is a numerical observation, not a gauge-invariant exponent. θ is a weak channel with no identifiable exponent: its free fit gives 1.024 over the full window and 1.67 on the 104-configuration subset, and its fitted amplitude is not predictive (0.012 at $\varepsilon = 1.2$ against a public median of 0.0228). For u_x and u_y the exponent is consistent across geometry (0.7 and 1.8 standard errors) and fluid (1.5%). Only the prefactor A must be calibrated, which is why the surrogate takes X as input rather than a table of A. A preliminary set of fifteen paired 3-D solutions at one Ra and one resolution reproduces the exponent (flow means 1.05–1.14, per-flow 1.03–1.15; Sec. 5.14).
3. A reason. The ε-expansion of the compressible member reproduces the Boussinesq system at $O(1)$ and produces the correction through $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$ and the density-modulated inertia. The linear operator of that system does not contain ε, so the correction factorises, $D = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots$, with a shape that does not rotate. All five consequences of the expansion listed in Table S1 are confirmed numerically.
4. A consistency check on the linearised operator. One Newton step of the retained low-Mach residual at X reproduces the measured correction to 4.7/3.8/7.2% (median, $u_x/u_y/p$) on the 27 samples of the $\varepsilon = 0.02$–0.10 sweep, where no extrapolation in ε is involved; θ is reproduced there to 12.2%, so its collapse at high contrast is an amplitude effect rather than a failure of the linearised operator. The extrapolated P also predicts held-out measurements at $\varepsilon = 0.1$ and 0.2 to 2.4/1.8/3.8%.
5. A design rule. Predicting $P(X)$ and multiplying by ε outside the network, which changes only the target and one multiplication, converts the law into a prior that (i) has 1.09× lower channel-averaged error than a FiLM-conditioned network in extrapolation (θ and p individually, p in the fixed gauge; the u_x/u_y intervals include zero), (ii) lets the network discover the ε^2 term itself, (iii) works the same way for a Fourier neural operator, and (iv) makes a public data set that exists at a single temperature contrast reusable at others—0.093 (u_x)

on $\varepsilon > 1.3$ against 0.144 for the plain model and 0.302 for doing nothing—which is the difference between a surrogate that must be retrained and one that can be applied. The scope of that reuse is set by Sec. 5.7: it holds within a boundary-condition family and fails across one, predictably.

The same factorisation lowers the mean u_x error of a POD-ROM with at most four modes on the 3-D samples from 0.168 to 0.081 over five held-out contrasts.

The heat-transfer audit adds a constraint that the field norm hides. At $\varepsilon = 1.2$ the public prior reduces the temperature-field error from 7.2% to 0.10% and the local wall-flux error from 6.0% to 4.3% at the hot wall, yet its integrated wall Nu error is 2.6% against 0.7% for doing nothing. The field norm does not see this bias. A relative L_1 penalty on the wall-normal gradient, added to the same training loss, brings the integrated error to 0.11% and the local error to 1.0%, so the failure is a property of the training target rather than of the factorisation. For the isothermal sidewall family the integral correction itself is small and changes sign between $\varepsilon = 1.2$ (+0.51%) and 1.9 (−1.89%).

Section 6 lists the limits: two dimensions for the main sweep, $\varepsilon < 2$, one-dimensional break symmetry, and a prefactor whose power-law fit leaves 16% (u_x), 18% (u_y) and 78% (θ) of the variance unexplained; the θ gap closes to 22% once the advection strength χ is supplied. The leading ε factor is derived; the amplitude, the window-averaged exponent and the ε^2 curvature are measured, and the curvature is also learned by the network.

8. Reproducibility

Table S7 of the supplementary material maps each script to the results it produces. The solver, surrogate and reporting scripts (dolfinx 0.7.3 for the solver, PyTorch 2.10 for the surrogates), together with the machine-readable results used in the tables and figures, are mirrored in the repository listed under Data availability. The raw solver output (600 random-boundary-condition, 104 aspect-2:1, 104 $Pr = 4.38$, 27 $\varepsilon \to 0$, 100 paired and 10 tight-rerun two-dimensional samples—the tight rerun repeats ten of the 600—plus ten 128^2 mesh-check pairs ($Ra = 10^4$ over five contrasts, $Ra = 10^5$ over two, $Ra = 10^6$ over three) and 15 paired three-dimensional samples (three flows × five contrasts), ∼6 GB) stays on the machine described above. The stored results are the source of every number in this paper; the reporting scripts regenerate them, and no number is transcribed by hand.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used an AI assistant (OpenAI Codex) to develop analysis scripts, to draft and edit the manuscript, and to check numerical consistency between the text and the stored results. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

Data availability

The public paired data set of Menglik et al. [5] is available under CC BY 4.0. Our generated data, solver configuration, and every script that reproduces the numbers in this paper are available at `https://github.com/sheep-cloud12138/Algorithms-instead-of-simulations` (branch `dev`); the raw solver output is available from the authors on request.

References

[1] M. Giacomini, P. Díez, Surrogates for physics-based and data-driven modelling of parametric systems: Review and new perspectives, Archives of Computational Methods in Engineering 33 (2026) 8519–8551. doi:10.1007/s11831-026-10552-4.

[2] C. Meng, S. Griesemer, D. Cao, S. Seo, Y. Liu, When physics meets machine learning: A survey of physics-informed machine learning, Machine Learning for Computational Science and Engineering 1 (2025) 20. doi:10.1007/s44379-025-00016-0.

[3] E. A. Spiegel, G. Veronis, On the Boussinesq approximation for a compressible fluid, The Astrophysical Journal 131 (1960) 442–447. doi:10.1086/146849.

[4] D. D. Gray, A. Giorgini, The validity of the Boussinesq approximation for liquids and gases, International Journal of Heat and Mass Transfer 19 (1976) 545–551. doi:10.1016/0017-9310(76)90168-X.

[5] N. Menglik, A. Shao, D. Hyde, A neural surrogate approach for simulating natural convection problems, 2026. arXiv:2606.25259.

[6] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, in: International Conference on Learning Representations, 2021. ArXiv:2010.08895.

[7] Q. Huhn, M. Tano, J. C. Ragusa, Y. Choi, Parametric dynamic mode decomposition for reduced order modeling, Journal of Computational Physics 475 (2022) 111852. doi:10.1016/j.jcp.2022.111852.

[8] F. Feng, Y. Li, Z. Chen, W.-T. Wu, J.-Z. Peng, M. Mei, Graph convolution network-based surrogate model for natural convection in annuli, Case Studies in Thermal Engineering 57 (2024) 104330. doi:10.1016/j.csite.2024.104330.

[9] J.-Z. Peng, N. Aubry, Y. Li, M. Mei, Z. Chen, W.-T. Wu, Physics-informed graph convolutional neural network for modeling geometry-adaptive steady-state natural convection, International Journal of Heat and Mass Transfer 216 (2023) 124593. doi:10.1016/j.ijheatmasstransfer.2023.124593.

[10] L. Lu, P. Jin, G. Pang, Z. Zhang, G. E. Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, Nature Machine Intelligence 3 (2021) 218–229. doi:10.1038/s42256-021-00302-5.

[11] S. Wang, H. Wang, P. Perdikaris, Learning the solution operator of parametric partial differential equations with physics-informed DeepONets, Science Advances 7 (2021) eabi8605. doi:10.1126/sciadv.abi8605.

[12] X. Zhao, X. Chen, Z. Gong, W. Zhou, W. Yao, Y. Zhang, RecFNO: A resolution-invariant flow and heat field reconstruction method from sparse observations via Fourier neural operator, International Journal of Thermal Sciences 195 (2023) 108619. doi:10.1016/j.ijthermalsci.2023.108619.

[13] X. F. Yu, W. Gao, K. Guo, Z. Zhang, H. Wang, M. Wang, S. Qiu, W. Tian, Fourier neural operator-based approach for real-time prediction of high-temperature heat pipes frozen startup, Applied Thermal Engineering 302 (2026) 131964. doi:10.1016/j.applthermaleng.2026.131964.

[14] I. Şahin, C. Moya, A. Mollaali, G. Lin, G. Paniagua, Deep operator learning-based surrogate models with uncertainty quantification for optimizing internal cooling channel rib profiles, International Journal of Heat and Mass Transfer 219 (2023) 124813. doi:10.1016/j.ijheatmasstransfer.2023.124813.

[15] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.045.

[16] S. Cai, Z. Wang, S. Wang, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks for heat transfer problems, ASME Journal of Heat and Mass Transfer 143 (2021) 060801. doi:10.1115/1.4050542.

[17] D. Jalili, S. Jang, M. Jadidi, G. Giustini, A. Keshmiri, Y. Mahmoudi, Physics-informed neural networks for heat transfer prediction in two-phase flows, International Journal of Heat and Mass Transfer 221 (2023) 125089. doi:10.1016/j.ijheatmasstransfer.2023.125089.

[18] T. De Ryck, S. Mishra, Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning, Acta Numerica 33 (2024) 633–713. doi:10.1017/S0962492923000089.

[19] K. Luo, J. Zhao, Y.-P. Wang, J. Li, J. Wen, J. Liang, H. Soekmadji, S. Liao, Physics-informed neural networks for PDE problems: A comprehensive review, Artificial Intelligence Review 58 (2025) 323. doi:10.1007/s10462-025-11322-7.

[20] P. K. Jha, Residual-based error corrector operator to enhance accuracy and reliability of neural operator surrogates of nonlinear variational boundary-value problems, Computer Methods in Applied Mechanics and Engineering 419 (2023) 116595. doi:10.1016/j.cma.2023.116595.

[21] L. Cao, T. O'Leary-Roseberry, P. K. Jha, J. T. Oden, O. N. Ghattas, Residual-based error correction for neural operator accelerated infinite-dimensional Bayesian inverse problems, Journal of Computational Physics 486 (2023) 112104. doi:10.1016/j.jcp.2023.112104.

[22] M. Zhu, H. Zhang, A. Jiao, G. E. Karniadakis, L. Lu, Reliable extrapolation of deep neural operators informed by physics or sparse observations, Computer Methods in Applied Mechanics and Engineering 412 (2023) 116064. doi:10.1016/j.cma.2023.116064.

[23] S. Goswami, K. Kontolati, M. D. Shields, G. E. Karniadakis, Deep transfer operator learning for partial differential equations under conditional shift, Nature Machine Intelligence 4 (2022) 1155–1164. doi:10.1038/s42256-022-00569-2.

[24] P. Conti, M. Guo, A. Manzoni, J. S. Hesthaven, Multi-fidelity surrogate modeling using long short-term memory networks, Computer Methods in Applied Mechanics and Engineering 404 (2022) 115811. doi:10.1016/j.cma.2022.115811.

[25] W. Chen, P. Stinis, Feature-adjacent multi-fidelity physics-informed machine learning for partial differential equations, Journal of Computational Physics 498 (2023) 112683. doi:10.1016/j.jcp.2023.112683.

[26] Y. Lyu, X. Zhao, Z. Gong, X. J. Kang, W. Yao, Multi-fidelity prediction of fluid flow based on transfer learning using Fourier neural operator, Physics of Fluids 35 (2023) 077118. doi:10.1063/5.0155555.

[27] G. Ahlers, S. Großmann, D. Lohse, Heat transfer and large scale dynamics in turbulent Rayleigh–Bénard convection, Reviews of Modern Physics 81 (2009) 503–537. doi:10.1103/RevModPhys.81.503.

[28] H. Yik, V. Valori, S. Weiss, Turbulent Rayleigh–Bénard convection under strong non-Oberbeck–Boussinesq conditions, Physical Review Fluids 5 (2020) 103502. doi:10.1103/PhysRevFluids.5.103502.

[29] M. Macek, G. Zinchenko, V. Musilová, P. Urban, J. Schumacher, Assessing non-Oberbeck–Boussinesq effects of convection in cryogenic helium, Physical Review Fluids 8 (2023) 094606. doi:10.1103/PhysRevFluids.8.094606.

[30] X. Pan, J.-I. Choi, Non-Oberbeck–Boussinesq effects on a water-filled differentially heated vertical cavity, Physics of Fluids 35 (2023) 113609. doi:10.1063/5.0173049.

[31] S.-N. Xia, Z.-H. Wan, S. Liu, Q. Wang, D.-J. Sun, Flow reversals in Rayleigh–Bénard convection with non-Oberbeck–Boussinesq effects, Journal of Fluid Mechanics 798 (2016) 628–642. doi:10.1017/jfm.2016.338.

[32] J. Zhang, Y. Wu, N. Gui, X. Yang, J. Tu, S. Jiang, Investigation of compressible flow under low Mach number in an enclosed square cavity with a novel non-Boussinesq algorithm, Physics of Fluids 35 (2023) 123120. doi:10.1063/5.0175738.

[33] H. Satake, T. Tagawa, Mathematical modeling for low-Mach-number compressible thermal convection analysis in a square enclosure, Physics of Fluids 36 (2024) 127168. doi:10.1063/5.0243786.

[34] P. Bella, E. Feireisl, F. Oschmann, Rigorous derivation of the Oberbeck–Boussinesq approximation revealing unexpected term, Communications in Mathematical Physics 403 (2023) 1245–1273. doi:10.1007/s00220-023-04823-5.

[35] Y. Song, Y. Wu, Y. Wang, X. Yang, DimINO: Dimension-informed neural operator learning, 2024. arXiv:2410.05894.